"Calibeating" Prediction-Powered Inference

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Abstract

Prediction-powered inference uses black-box predictions to improve estimation from small labeled samples and large unlabeled samples, but raw prediction scores are often miscalibrated and can be inefficient regression adjustments. We study semisupervised mean estimation with a black-box score, a small labeled sample, and a large unlabeled sample. Augmented inverse-probability weighting (AIPW) and PPI give valid inference for fixed or cross-fitted scores, but their efficiency depends on how well the score predicts the outcome. We propose *calibrated prediction-powered inference*: post-hoc calibrate the score on the labeled sample, then average the calibrated predictions over the pooled covariate sample. The estimator requires no retraining, takes a simple plug-in form, and has an exact AIPW representation. For linear calibration, we show first-order equivalence to PPI++. For isotonic calibration, we establish asymptotic normality, valid Wald inference, and "*calibeating*" guarantees: isotonic post-processing improves the score as a predictor and as a first-order regression adjustment within the monotone class, and subsequent score-only post-processing yields no additional first-order gain. We also show that the original PPI estimator is a special case of AIPW and can be inefficient when the prediction score is already accurate. Simulations, benchmark reproductions, and an LLM-evaluation application show that calibrated estimators often improve on PPI and are competitive with AIPW and PPI++.

1 Introduction

We consider semisupervised mean estimation with one labeled dataset $\{(X_i, Y_i)\}_{i=1}^n$ and one unlabeled dataset $\{\widetilde{X}_j\}_{j=1}^N$, where outcomes are observed only for a subset sampled completely at random. This is a classical missing-outcome problem: covariates are available for many individuals, but the outcome of interest is observed only for a smaller subset. Such settings arise when gold-standard outcomes are expensive to collect but prediction scores are readily available, for example in randomized trials, biobank analyses, and model-assisted scientific pipelines [89, 43, 115, 1, 118].

The goal is to use the larger unlabeled sample to improve estimation of a population mean while accounting for prediction error. This problem has long been studied in missing-data and semiparametric statistics, where AIPW, TMLE, and related debiased estimators combine flexible prediction with valid inference [75, 103, 105, 104, 18]. Closely related ideas have recently appeared in the machine-learning literature as prediction-powered inference [1, 2, 118]. Similar ideas also arise in covariate adjustment for randomized trials, where machine-learning predictors trained on auxiliary data can improve precision in small samples [35, 83, 61, 58, 89, 4, 42].

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A standard approach is to debias a plug-in estimator based on a prediction score $m(X)$. The AIPW estimator averages $m(X)$ over the pooled covariate sample and adds the average labeled residual $Y - m(X)$ [75]. The original PPI estimator instead averages $m(X)$ only over the unlabeled sample [1]. Both estimators remain valid when m is misspecified, but their efficiency depends on how well $m(X)$ serves as a regression adjustment. A score may rank observations well while being poorly calibrated, so its numerical values can remain misaligned with the conditional mean of Y.

This gap motivates post-hoc calibration. Rather than correcting only the marginal bias $\mathbb{E}\{Y - m(X)\}$, we use the labeled sample to transform $m(X)$ so that its values better match the outcome scale [113, 114, 65]. Ideally, the transformed score $q(X)$ satisfies $q(X) \approx \mathbb{E}\{Y \mid q(X)\}$: among observations with the same calibrated value, the average outcome is close to that value. We calibrate the black-box score on the labeled sample and average the calibrated predictions over the pooled covariate sample. We call this procedure *calibrated prediction-powered inference*. Calibration can improve both predictive accuracy and the efficiency of the resulting mean estimator.

Contributions. Our contributions are threefold. First, we develop a calibrated-PPI framework that includes standard post-hoc rules such as linear [60, 53], histogram binning [113, 30], and isotonic calibration [114, 65]. These procedures can be added to an existing prediction pipeline without retraining the original score. Second, we show that the resulting plug-in estimator is exactly an AIPW estimator based on the calibrated score. This identity explains why calibration can improve efficiency without changing validity: calibration changes the regression adjustment, while the estimator retains its AIPW form. Third, we establish asymptotic theory for the main calibrated estimators. Linearly calibrated PPI is first-order equivalent to unconstrained PPI++, while isotonic calibration yields asymptotic linearity and valid Wald inference. Isotonic calibration achieves the smallest asymptotic variance among nondecreasing transformations of the original score. Further post-processing of the fitted isotonic score yields no additional first-order variance reduction. Following the terminology of Foster and Hart [25], we call improvement through calibration *calibeating*. Here, these improvements concern squared prediction error and the first-order variance of semisupervised mean estimation.

1.1 Related work

Several recent PPI estimators can be viewed as AIPW estimators with different choices of regression-adjustment score. Cross-PPI uses cross-fitting to avoid overfitting bias, while PPI++ chooses an empirical scaling of the score to improve efficiency. This view connects PPI to classical missing-data, semiparametric, debiased machine-learning, and covariate-adjustment methods for mean estimation [75, 103, 105, 104, 18, 89, 42]. Semiparametric efficiency theory shows that precision depends on the quality of the regression-adjustment score, while unbiasedness does not require the score to be correctly specified [75, 5, 74, 112].

The closest precursor to our approach is calibrated debiased machine learning, which calibrates outcome-regression and debiasing nuisance estimators inside AIPW procedures [100]. That work targets robustness to nuisance misspecification in nonrandomized settings; here, random sampling already gives unbiasedness for any fixed score, so calibration is used to improve efficiency and justify post-processing a fixed score. Recalibrated prediction-powered inference (RePPI) modifies the estimating equation with a flexible machine-learned correction to target full semiparametric efficiency [45]. In contrast, we post-process the fixed prediction score using standard calibration maps and characterize the resulting guarantees for semisupervised mean estimation. Appendix B gives the full related-work discussion.

2 Two-Sample Semisupervised Setup

2.1 Data structure and notation.

We consider the standard prediction-powered inference setting with a small labeled sample and a larger unlabeled sample:

$$\mathcal{D}_L = \{(X_i, Y_i)\}_{i=1}^{n}, \qquad \mathcal{D}_U = \{\widetilde{X}_j\}_{j=1}^{N}.$$

The labeled observations are i.i.d. from P_0 on $\mathcal{X} \times \mathbb{R}$, while the unlabeled observations are i.i.d. from the X-marginal $P_{0,X}$ of P_0. The estimand is the population mean

$$\psi_0 := \mathbb{E}_{P_0}[Y], \qquad \mu_0(x) := \mathbb{E}_{P_0}[Y \mid X = x].$$

Let $m : \mathcal{X} \to \mathbb{R}$ denote a black-box prediction model, trained externally or on an independent split, and write $T = m(X)$ for its scalar prediction.

For functions $a : \mathcal{X} \times \mathbb{R} \to \mathbb{R}$ and $b : \mathcal{X} \to \mathbb{R}$, define the empirical means over the labeled and unlabeled samples by

$$\mathbb{P}_n^L a := \frac{1}{n}\sum_{i=1}^{n} a(X_i, Y_i), \qquad \mathbb{P}_N^U b := \frac{1}{N}\sum_{j=1}^{N} b(\widetilde{X}_j).$$

Let $M := n + N$ and $\rho_n := n/M$. For $q : \mathcal{X} \to \mathbb{R}$, define the pooled covariate empirical mean

$$\mathbb{P}_M^X q := \frac{1}{M}\Big\{\sum_{i=1}^{n} q(X_i) + \sum_{j=1}^{N} q(\widetilde{X}_j)\Big\} = \rho_n \mathbb{P}_n^L\{q(X)\} + (1-\rho_n)\mathbb{P}_N^U\{q(\widetilde{X})\}.$$

Assumption 1 (Two-sample design)**.** The following hold:

(i) $\mathcal{D}_L$ is i.i.d. from P_0, $\mathcal{D}_U$ is i.i.d. from $P_{0,X}$, and the two samples are independent.

(ii) Y has finite second moment under P_0.

(iii) As $n \to \infty$, the labeled fraction $\rho_n = n/(n+N)$ satisfies $\rho_n \to \rho_0$ for some $\rho_0 \in [0, 1)$.

2.2 Review of AIPW, PPI, and semiparametric efficiency

We first review the estimator class and efficiency comparison that motivate score calibration.

The AIPW class. For any regression-adjustment score $q : \mathcal{X} \to \mathbb{R}$ with finite second moment, define

$$\widehat{\psi}(q) := \mathbb{P}_M^X q + \mathbb{P}_n^L\{Y - q(X)\}. \tag{1}$$

This is the AIPW estimator in the unrestricted two-sample model [75]. For any fixed q, or for an estimated score constructed independently of the labeled outcomes used in the residual term, $\mathbb{E}\{\widehat{\psi}(q) \mid q\} = \psi_0$; thus the score affects efficiency but not unbiasedness.

Asymptotic variance and efficiency. Appendix C gives the asymptotic expansion of $\widehat{\psi}(q)$. Its main consequence is

$$\mathrm{AVar}\Big\{\sqrt{n}\,(\widehat{\psi}(q) - \psi_0)\Big\} = V_{\mathrm{eff}} + (1-\rho_0)\mathbb{E}_{P_0}\left[(q^\circ(X) - \mu_0(X))^2\right],$$

where $q^\circ(X) := q(X) - \mathbb{E}_{P_{0,X}}\{q(X)\} + \psi_0$ is the recentered score and $V_{\mathrm{eff}} = \rho_0 \mathrm{Var}\{\mu_0(X)\} + \mathbb{E}[\mathrm{Var}(Y \mid X)]$ is the semiparametric efficiency bound [8]. Hence efficiency is governed by how well q° approximates μ_0, and is optimized when $q^\circ = \mu_0$.

AIPW, PPI, and empirical-efficiency maximization. The usual AIPW estimator uses $q = m$. The original PPI estimator of Angelopoulos et al. [1] corresponds to the score $q = m/(1-\rho_n)$, since

$$\widehat{\psi}_{\mathrm{PPI}} = \mathbb{P}_N^U\{m(\widetilde{X})\} + \mathbb{P}_n^L\{Y - m(X)\}.$$

Thus AIPW and PPI are two choices of score inside the same AIPW class. Relative to AIPW with score m, PPI omits the labeled-sample covariate average from the plug-in term, which can reduce efficiency when the raw score is already a good regression adjustment.

Because both estimators lie in the scaling class $\{\lambda m(X) : \lambda \in \mathbb{R}\}$, one can choose λ by minimizing an empirical estimate of the AIPW variance. This is empirical efficiency maximization [83], and yields PPI++ [2]: the AIPW estimator with score $\hat{\lambda} m(X)$, targeting the optimal scaling $\lambda^\star = \frac{\mathrm{Cov}\{Y, m(X)\}}{\mathrm{Var}\{m(X)\}}$. The same idea can be applied to richer score classes, such as linear augmentations, monotone transformations, or RKHS classes [83, 103]; see Appendix D. This motivates calibration: keep the AIPW form fixed, and improve the score used inside it.

3 Calibrated Prediction-Powered Inference

The previous section shows why calibration matters: AIPW gives unbiasedness for any fixed score, but efficiency depends on how well that score approximates the outcome regression. Standard AIPW and prediction-powered estimators use the raw prediction score $m(X)$. In applications, this score may rank individuals well while its numerical scale remains poorly aligned with the outcome.

Calibration addresses this problem by transforming $m(X)$ into a new score $q(X)$ whose values better match the outcome scale. Ideally,

$$q(X) \approx \mathbb{E}_{P_0}\{Y \mid q(X)\},$$

so that, among individuals assigned the same calibrated value, the average outcome is close to that value. Under squared loss, the best score-only post-processing of the original prediction is

$$q_0^{\mathrm{score}}(X) := \mathbb{E}_{P_0}\{Y \mid m(X)\}.$$

Thus calibration can be viewed as moving $m(X)$ toward $q_0^{\mathrm{score}}(X)$ without retraining the underlying model.

We study a simple post-hoc strategy: calibrate $m(X)$ on the labeled sample, then plug the calibrated score into the semisupervised mean estimator. We call the resulting approach *calibrated prediction-powered inference*. It gives a plug-in estimator with an exact AIPW representation. Different calibration classes yield different calibrated PPI estimators, including the linear and isotonic procedures studied below.

3.1 General estimator and exact AIPW representation

Let $m : \mathcal{X} \to \mathbb{R}$ be a black-box score trained on external data or on a separate training sample, and write $T = m(X)$. We estimate a one-dimensional calibration map $g_n : \mathbb{R} \to \mathbb{R}$ from the labeled sample and replace m by

$$q_n := g_n \circ m.$$

Thus $q_n(X)$ is a post-processed version of the original score whose scale is chosen to better match the observed outcomes.

The calibrated plug-in estimator averages this calibrated score over all available covariates:

$$\widehat{\psi} := \mathbb{P}_M^X q_n = \frac{1}{M}\Big\{\sum_{i=1}^{n} q_n(X_i) + \sum_{j=1}^{N} q_n(\widetilde{X}_j)\Big\}. \tag{2}$$

Let $\mathcal{H}$ be a class of post-processing maps $g : \mathbb{R} \to \mathbb{R}$. We fit g_n on the labeled sample by least squares:

$$g_n \in \arg\min_{g \in \mathcal{H}} \mathbb{P}_n^L\big[(Y - g(T))^2\big], \tag{3}$$

and write $q_n = g_n \circ m$. If $\mathcal{H}$ includes an intercept, or is closed under additive constants, the least-squares normal equations give

$$\mathbb{P}_n^L\{Y - q_n(X)\} = 0.$$

Hence the calibrated plug-in estimator is exactly the AIPW estimator with score q_n:

$$\widehat{\psi}_{\mathrm{cal}} = \mathbb{P}_M^X q_n + \mathbb{P}_n^L\{Y - q_n(X)\}. \tag{4}$$

The exact AIPW representation does not imply exact unbiasedness when q_n is fitted using the same labeled outcomes. Because calibration uses only the labeled sample, conditioning on that sample gives, whenever the expectations are finite,

$$\mathbb{E}(\widehat{\psi}_{\mathrm{cal}} - \psi_0) = -(1 - \rho_n)\mathbb{E}\big[(\mathbb{P}_n^L - P_{0,X})q_n\big].$$

The right-hand side is zero for a fixed score, but need not vanish for a fitted score. The results below establish asymptotic linearity for linear and isotonic calibration without sample splitting. Appendix D.1 gives exactly unbiased cross-fitted and leave-one-out AIPW estimators that retain the residual correction and exclude each observation from its calibration fit. The main examples below are linear and isotonic calibration, corresponding to affine and monotone choices of $\mathcal{H}$. Thus post-hoc calibration changes the score, not the AIPW form of the estimator.

Why calibration can improve efficiency. For a fixed score q, the asymptotic variance of $\sqrt{n}\{\widehat{\psi}(q) - \psi_0\}$ can also be written as

$$V(q) = \rho_0 \mathrm{Var}(Y) + (1 - \rho_0)\mathrm{Var}\{Y - q(X)\}.$$

Suppose $\mathcal{H}$ is closed under addition of constants, and let g_0 minimize $\mathbb{E}[\{Y - g(T)\}^2]$ over $g \in \mathcal{H}$. Because adding a constant to a score leaves $V(q)$ unchanged, $q_0 = g_0 \circ m$ also minimizes the asymptotic variance over the same class of transformed scores. If the identity map belongs to $\mathcal{H}$, then $V(q_0) \leq V(m)$. If $\mathcal{H}$ contains $t \mapsto \mathbb{E}(Y \mid T = t)$, then $q_0(X) = \mathbb{E}(Y \mid T)$, and AIPW with this score attains the efficiency bound in the model based on (T, Y).

Under the conditions below, linear and isotonic calibration have the same first-order variance as their population counterparts. Affine calibration estimates an intercept and slope; isotonic calibration allows any nondecreasing transformation. The choice determines both the reduction in population prediction error and the variability of estimating the calibration map from limited labeled data.

3.2 Isotonic-calibrated PPI

Our main flexible procedure is isotonic calibration, a nonparametric post-hoc calibration method [113, 114, 65]. Let $\mathcal{F}_{\mathrm{iso}}$ be the class of nondecreasing maps. Writing $T_i = m(X_i)$, we fit

$$\hat{g}_n^{\mathrm{iso}} \in \arg\min_{g \in \mathcal{F}_{\mathrm{iso}}} \sum_{i=1}^{n} \{Y_i - g(T_i)\}^2, \qquad q_n^{\mathrm{iso}}(X) := \hat{g}_n^{\mathrm{iso}}\{m(X)\},$$

and define

$$\widehat{\psi}_{\mathrm{iso}} := \mathbb{P}_M^X q_n^{\mathrm{iso}}.$$

Isotonic calibration is useful when the score ranks observations well but its scale is distorted. It preserves the score ordering up to ties while choosing the empirical risk-minimizing monotone transformation of the predicted values. Because the identity map is monotone, it cannot fit worse on the labeled sample than the raw score. Isotonic regression requires no tuning and is widely available in standard software. For new score values, we interpolate linearly between the fitted values at distinct observed scores and use the endpoint values outside the observed range.

Proposition 1 (Isotonic calibration is empirically calibrated)**.** *For every measurable $h : \mathbb{R} \to \mathbb{R}$,*

$$\mathbb{P}_n^L\big[h(q_n^{\mathrm{iso}}(X))\{Y - q_n^{\mathrm{iso}}(X)\}\big] = 0. \tag{5}$$

Moreover, for every $\theta \in \mathcal{F}_{\mathrm{iso}}$,

$$\mathbb{P}_n^L\big[(Y - q_n^{\mathrm{iso}}(X))^2\big] \leq \mathbb{P}_n^L\big[(Y - \theta(m(X)))^2\big].$$

In particular, since the identity map is monotone, isotonic calibration fits no worse on the labeled sample than the raw score m.

The orthogonality holds because the fitted isotonic score is constant on the blocks learned by the pooled-adjacent-violators algorithm [6], and each block value equals the average outcome in that block [98]. Therefore any function of the fitted score is also constant on these blocks, making the residuals orthogonal to all score-only post-processing functions. Histogram binning, scalable monotone implementations, and cross-fitted variants are discussed in Appendices D and O.

3.3 Linearly calibrated PPI

When the labeled sample is small, affine calibration can be more stable than isotonic regression. It fits only a shift and rescaling of the score, so it cannot repair general nonlinear miscalibration. It also matches the familiar use of a prognostic score in ordinary regression adjustment. We fit

$$(\hat{a}_{\mathrm{lin}}, \hat{b}_{\mathrm{lin}}) \in \arg\min_{a,b \in \mathbb{R}} \sum_{i=1}^{n} \{Y_i - a\, m(X_i) - b\}^2, \qquad q_n^{\mathrm{lin}}(X) := \hat{a}_{\mathrm{lin}}\, m(X) + \hat{b}_{\mathrm{lin}}. \tag{6}$$

The corresponding plug-in estimator is

$$\widehat{\psi}_{\mathrm{lin}} := \mathbb{P}_M^X q_n^{\mathrm{lin}}. \tag{7}$$

After linear calibration, no further shift or rescaling of the score can reduce the labeled-sample squared error.

The least-squares normal equations give $\mathbb{P}_n^L\{Y - q_n^{\mathrm{lin}}(X)\} = 0$ and $\mathbb{P}_n^L[q_n^{\mathrm{lin}}(X)\{Y - q_n^{\mathrm{lin}}(X)\}] = 0$. Thus every affine weight $a + bq_n^{\mathrm{lin}}$ gives a zero empirical residual correction in the AIPW representation.

The next result connects linear calibration to PPI++ [2]. The variance-minimizing coefficient and the least-squares slope converge to the same population coefficient. Let $\hat{\lambda}_{++}$ minimize the empirical AIPW variance over $\lambda \in \mathbb{R}$, and write $\widehat{\psi}_{++}(\hat{\lambda}_{++}) := \widehat{\psi}(\hat{\lambda}_{++}m)$.

Proposition 2 (First-order equivalence of PPI++ and linear calibration)**.** *Assume Assumption 1,* $\mathrm{Var}\{m(X)\} > 0$*, and* $\mathbb{E}_{P_0}\{m(X)^2\} < \infty$*. Then*

$$\widehat{\psi}_{++}(\hat{\lambda}_{++}) - \widehat{\psi}_{\mathrm{lin}} = (1 - \rho_n)(\hat{\lambda}_{++} - \hat{a}_{\mathrm{lin}})\{\mathbb{P}_N^U m - \mathbb{P}_n^L m\}.$$

Moreover, $\hat{\lambda}_{++} - \hat{a}_{\mathrm{lin}} = o_p(1)$*, and hence*

$$\widehat{\psi}_{++}(\hat{\lambda}_{++}) - \widehat{\psi}_{\mathrm{lin}} = o_p(n^{-1/2}).$$

Thus empirical efficiency maximization over $\lambda \in \mathbb{R}$ yields the same first-order estimator as linearly calibrated PPI, but the calibration view gives the procedure the familiar interpretation of ordinary regression adjustment. Implementations that clip or constrain λ need not have this equivalence [2]. Platt scaling, covariate-augmented linear calibration, and cross-fitting are discussed in Appendix D.

4 Theory for isotonic-calibrated PPI

We next give first-order theory for isotonic calibration. The results formalize "calibeating" in two senses. First, isotonic calibration improves prediction within the class of monotone post-processing rules for the original score and improves first-order efficiency for mean estimation. Second, once the isotonic score has been fit, further post-processing of that score yields no nonvanishing reduction in squared prediction error or first-order reduction in variance.

4.1 Asymptotic normality, inference, and efficiency

Let g_0^{iso} be the population isotonic calibration map for the fixed score m:

$$g_0^{\mathrm{iso}} \in \arg\min_{g \in \mathcal{F}_{\mathrm{iso}}} \mathbb{E}_{P_0}\big[\{Y - g(T)\}^2\big], \qquad q_0^{\mathrm{iso}} := g_0^{\mathrm{iso}} \circ m. \tag{8}$$

Thus q_0^{iso} is the best monotone post-processing of the score under squared loss.

Assumption 2 (Regularity for isotonic calibration)**.** The following conditions hold.

(i) **Boundedness.** There exists $C_0 < \infty$ such that $|\mathbb{E}_{P_0}[Y \mid m(X)]| \le C_0$ almost surely, and $\sup_x |q_n^{\mathrm{iso}}(x)| = O_p(1)$.

(ii) **Tails.** For $T = m(X)$, there exists $0 < K < \infty$ such that

$$\mathbb{E}_{P_0}\left[\exp\left\{\frac{|Y - \mathbb{E}_{P_0}[Y \mid T]|}{K}\right\} \middle| T\right] \le 2 \qquad \text{almost surely.}$$

Theorem 1 (Asymptotic linearity under the two-sample model)**.** *Suppose Assumptions 1 and 2 hold. Then*

$$\|q_n^{\mathrm{iso}} - q_0^{\mathrm{iso}}\|_{2,P_{0,X}}^2 = O_p(n^{-2/3}).$$

Define

$$D^L_{q_0^{\mathrm{iso}},n}(X,Y) := Y - \psi_0 - (1-\rho_n)\{q_0^{\mathrm{iso}}(X) - \psi_0\}, \qquad D^U_{q_0^{\mathrm{iso}},n}(\widetilde{X}) := (1-\rho_n)\{q_0^{\mathrm{iso}}(\widetilde{X}) - \psi_0\}.$$

Then

$$\widehat{\psi}_{\mathrm{iso}} - \psi_0 = (\mathbb{P}_n^L - P_0)D^L_{q_0^{\mathrm{iso}},n} + (\mathbb{P}_N^U - P_{0,X})D^U_{q_0^{\mathrm{iso}},n} + R_{n,N}, \tag{9}$$

$$R_{n,N} = O_p\big((1-\rho_n)n^{-2/3}\big) + O_p\big((1-\rho_n)n^{-1/3}N^{-1/2}\big) = o_p(n^{-1/2}).$$

Consequently, $\widehat{\psi}_{\mathrm{iso}}$ *is asymptotically linear at the* $\sqrt{n}$ *rate with triangular influence pair* $(D^L_{q_0^{\mathrm{iso}},n}, D^U_{q_0^{\mathrm{iso}},n})$.

The asymptotic linear representation yields a mean-zero normal limit for $\sqrt{n}(\widehat{\psi}_{\text{iso}} - \psi_0)$ with variance

$$\sigma_0^2 := \text{Var}\big[Y - \psi_0 - (1-\rho_0)\{q_0^{\text{iso}}(X) - \psi_0\}\big] + \rho_0(1-\rho_0)\text{Var}\{q_0^{\text{iso}}(X)\}.$$

The following corollary establishes asymptotic normality and valid Wald inference using an empirical estimate of this variance.

Corollary 1 (Asymptotic normality and Wald inference)**.** *Under the conditions of theorem 1, assume that $\sigma_0^2 > 0$. Define the empirical variance estimator*

$$\widehat{\sigma}^2 := \mathbb{P}_n^L\{(\widehat{D}^L - \mathbb{P}_n^L\widehat{D}^L)^2\} + \frac{n}{N}\mathbb{P}_N^U\{(\widehat{D}^U - \mathbb{P}_N^U\widehat{D}^U)^2\},$$

where

$$\widehat{D}_i^L := Y_i - \widehat{\psi}_{\text{iso}} - (1-\rho_n)\{q_n^{\text{iso}}(X_i) - \widehat{\psi}_{\text{iso}}\}, \qquad \widehat{D}_j^U := (1-\rho_n)\{q_n^{\text{iso}}(\widetilde{X}_j) - \widehat{\psi}_{\text{iso}}\}.$$

Then

$$\frac{\sqrt{n}\,(\widehat{\psi}_{\text{iso}} - \psi_0)}{\widehat{\sigma}} \xrightarrow{d} N(0,1).$$

Corollary 2 (Efficiency)**.** *Under the conditions of theorem 1, the following hold.*

(i) If $q_0^{\text{iso}} = \mu_0$ almost surely, then $\widehat{\psi}_{\text{iso}}$ is efficient in the full two-sample model.

(ii) For any nondecreasing $\theta : \mathbb{R} \to \mathbb{R}$ such that $\theta \circ m$ is square-integrable, $\widehat{\psi}_{\text{iso}}$ has asymptotic variance no larger than the AIPW estimator with score $\theta \circ m$.

(iii) If $\mathbb{E}_{P_0}[Y \mid m(X)]$ is nondecreasing in $m(X)$, then

$$q_0^{\text{iso}}(X) = \mathbb{E}_{P_0}[Y \mid m(X)] \quad \textit{almost surely}.$$

In this case, no AIPW estimator based on a square-integrable transformation $\theta \circ m$ has smaller asymptotic variance. Equivalently, $\widehat{\psi}_{\text{iso}}$ is efficient in the reduced two-sample model that observes the scalar score $m(X)$ instead of X.

Taking θ constant in part (ii) shows that isotonic calibration is asymptotically at least as efficient as the labeled mean. At the $\sqrt{n}$ scale, the variance reduction is

$$\text{Var}(Y) - \sigma_0^2 = (1-\rho_0)\text{Var}\{q_0^{\text{iso}}(X)\}.$$

Isotonic calibration is fully efficient only if the score contains enough information to recover $\mu_0(X)$. Even when the score is only a one-dimensional summary, isotonic calibration is first-order at least as efficient as any nondecreasing post-processing of that score. This includes AIPW with the raw score, PPI, linear calibration with a nonnegative slope, and PPI++ when its fitted scaling coefficient is nonnegative. If $\mathbb{E}_{P_0}[Y \mid m(X)]$ is nondecreasing in $m(X)$, isotonic calibration is first-order optimal among all score-only post-processing rules.

When $N/n \to \infty$, the limiting labeled fraction is zero, and the asymptotic variance reduces to

$$\sigma_0^2 = \text{Var}\{Y - q_0^{\text{iso}}(X)\}.$$

If the conditional mean of Y given $m(X)$ is nondecreasing in the score, this equals $\mathbb{E}[\text{Var}\{Y \mid m(X)\}]$. The unlabeled contribution is negligible at the $\sqrt{n}$ scale, and the labeled residuals determine first-order uncertainty.

4.2 No further first-order gain from post-hoc calibration

The previous result compares isotonic calibration with monotone transformations of the original score. We now ask whether, after fitting the isotonic score $q_n^{\text{iso}}(X)$, any additional score-only post-processing can still improve first-order inference. Since $\widehat{\psi}_{\text{iso}}$ uses the covariates only through $q_n^{\text{iso}}(X)$, the relevant comparison is the reduced two-sample experiment

$$\Big(\{(q_n^{\text{iso}}(X_i), Y_i)\}_{i=1}^n,\ \{q_n^{\text{iso}}(\widetilde{X}_j)\}_{j=1}^N\Big).$$

In this reduced experiment, the best possible score-only post-processing is

$$q_{0|n}(X) := \mathbb{E}_{P_0}\{Y \mid q_n^{\text{iso}}(X)\}.$$

Assumption 3 (Bounded-variation score regression)**.** The map $t \mapsto \mathbb{E}_{P_0}[Y \mid m(X) = t]$ has bounded variation on the support of $m(X)$.

The bounded-variation assumption permits a nonmonotone conditional mean, including functions with jumps. It does not require differentiability or Lipschitz continuity.

Theorem 2 (No further first-order gain from post-hoc calibration)**.** *Suppose Assumptions 1 to 3 hold. Let*

$$\widehat{\psi}_{\text{eff}} := \widehat{\psi}(q_{0|n}), \qquad q_{0|n}(X) := \mathbb{E}_{P_0}\{Y \mid q_n^{\text{iso}}(X)\}.$$

Then

$$\|q_{0|n} - q_n^{\text{iso}}\|_{2,P_{0,X}}^2 = O_p(n^{-2/3}), \qquad \widehat{\psi}_{\text{eff}} - \widehat{\psi}_{\text{iso}} = o_p(n^{-1/2}).$$

Thus $\widehat{\psi}_{\text{iso}}$ and $\widehat{\psi}_{\text{eff}}$ have the same asymptotic variance.

Corollary 2 compares nondecreasing transformations of the original score. Theorem 2 considers arbitrary measurable transformations applied after isotonic calibration. The conditional expectation defining $q_{0|n}$ is evaluated for a new observation from P_0, with the fitted isotonic map held fixed. Replacing the fitted score by this population conditional mean changes the mean estimator by $o_p(n^{-1/2})$. Thus, further calibration of the fitted isotonic score cannot reduce its first-order asymptotic variance.

5 Experiments

We compare the labeled-only mean, AIPW, PPI, PPI++, and AIPW-EM with `LinearCal`, `MonoSpline`, `IsoCal`, and `AutoCal`. Linear calibration fits an affine map, monotone spline calibration fits a smooth nondecreasing function, and isotonic calibration fits a nondecreasing function. AutoCal selects among AIPW and these calibration methods using cross-validated estimates of the AIPW variance. In the real-data and LLM benchmarks, LinearCal clips affine predictions to the labeled outcome range, and IsoCal first groups observations into bins of at least 10 before fitting isotonic regression; Appendix N gives the fitting details.

We first examine score miscalibration and small labeled samples in simulations, then evaluate the methods on public PPI benchmarks and LLM evaluation tasks. We report bias, Wald-interval coverage, MSE, and Monte Carlo standard deviation. Empirical relative efficiency is the ratio of the reference estimator's Monte Carlo variance to that of the estimator being evaluated; each caption specifies the reference estimator.

Simulation study We generate $S \sim N(0,1)$ and $Y \mid S \sim \text{Bernoulli}\{\mu_0(S)\}$, where $\mu_0(S) = \sigma(5S)$ and $\sigma(t) = (1+e^{-t})^{-1}$. The prediction score is

$$m(X) = C[-0.15 + 0.75\,\sigma\{3\,\text{logit}\{\mu_0(S)\} - 0.5\}]\,,$$

where $C(t) = \min\{0.99, \max(0.01, t)\}$. This transformation preserves the ordering of the conditional mean, apart from ties induced by clipping, while introducing miscalibration on the probability scale. We consider $n \in \{50, 100, 200, 400, 800, 1200, 2400\}$, with $N/n \in \{1, 16\}$, and use 500 Monte Carlo replications per setting.

At $n = 400$ and $N/n = 16$, `LinearCal`, `MonoSpline`, and `IsoCal` have RMSEs between 0.0164 and 0.0166, compared with 0.0172 for PPI and 0.0175 for AIPW. Their Wald-interval coverage ranges from 0.910 to 0.932.

The PPI++ implementation restricts the scaling coefficient to $[0, 1]$ in the original PPI parameterization, equivalent to $[0, (1-\rho_n)^{-1}]$ in the AIPW parameterization. AIPW-EM minimizes the same empirical variance criterion without this constraint and performs similarly to `LinearCal`. Averaged across sample sizes with $N/n = 16$, RMSE is approximately 0.0208 for PPI++, compared with 0.0196 for AIPW-EM and `LinearCal`.

Small-sample comparisons We also examine smaller labeled samples under the same binary-outcome model, using mild and strong score miscalibration as specified in Appendix N.1. We set $N = 16n$ for $n \in \{20, 50, 100, 200, 400, 800, 1200\}$ and use 1,000 replications per setting. The labeled mean, PPI, AIPW, PPI++, linear, smooth monotone, and isotonic calibration are evaluated on the same samples, together with the leave-one-out linear and isotonic estimators in (14).

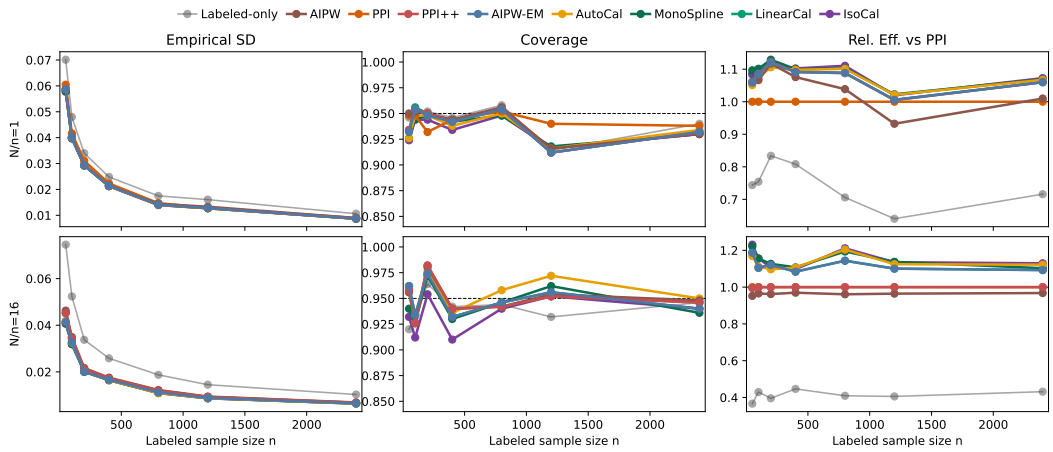


Figure 1: Synthetic simulation study with a poorly calibrated score. The top row shows $N/n = 1$, the bottom row shows $N/n = 16$. Columns report Monte Carlo standard deviation, coverage, and relative efficiency versus PPI.

Under strong miscalibration, linear, smooth monotone, and isotonic calibration each have lower RMSE than PPI++ at all seven sample sizes. At $n = 20$, isotonic coverage ranges from 0.814 to 0.871 across the two settings; leave-one-out calibration increases it to 0.905–0.925, with wider intervals and larger RMSE. Across all 14 settings, leave-one-out calibration lowers linear-calibration RMSE by at most 1.99%, but lowers isotonic RMSE in only one setting. The RePPI random-forest and neural-network comparisons are reported in Appendix N.1.

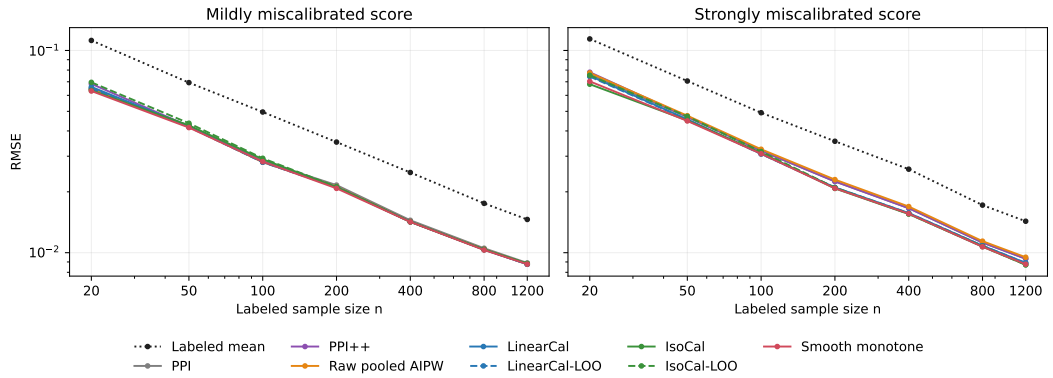


Figure 2: RMSE across labeled sample sizes under mild and strong score miscalibration, with $N = 16n$ and 1,000 replications per setting. Solid and dashed curves distinguish calibration on the labeled sample from leave-one-out calibration.

Reproduced PPI benchmarks We next evaluate the same idea on the standard PPI mean-estimation benchmarks of [1], where any score miscalibration is present in the public score rather than introduced by design. Using the public `ppi_py` `forest`, `galaxies`, and `census_income` datasets, we rerandomize the labeled/unlabeled split 500 times at each labeled-sample size and evaluate all estimators on the same splits.

AIPW is usually the strongest raw-score baseline; PPI is competitive mainly when the unlabeled sample is extremely large. Calibration gains are therefore not uniform. On `forest` and `galaxies`, the leading score-based methods are tightly clustered. On `census_income`, the score is sufficiently mis-scaled that `LinearCal` has the lowest average MSE. The efficiency-maximization comparison also matches the theory: across the 30 combinations of dataset and labeled sample size, unclipped AIPW-EM has lower MSE than clipped PPI++ in 29 combinations and is nearly tied with

`LinearCal`. This indicates that much of the observed PPI++ gap is due to clipping rather than empirical efficiency maximization itself. Appendix N gives the dataset-by-regime tables and the full calibration comparison.

LLM evaluation benchmark The final experiment studies modern LLM evaluation, where proxy-score miscalibration is common and raw reward-model margins can be badly scaled. The first track uses PPE Human Preference V1 [26]; the second uses a PPE correctness suite built from MMLU-Pro, MATH, GPQA, IFEval, and MBPP+ [26, 110, 39, 73, 117, 59]. In PPE Human, the estimand for each target model is its human win rate, counting ties as half-wins. In PPE Correctness, each observation is a pair containing one correct and one incorrect response from the same model, and the estimand is the probability that the first response is correct. We use four public precomputed evaluator-score sources on the PPE tracks: Skywork-Reward-Llama-3.1-8B, ArmoRM-Llama3-8B-v0.1, Athene-RM-8B, and Starling-RM-7B-alpha. We use 200 random labeled/unlabeled splits for each combination of evaluator, benchmark, and target model.

On PPE Human at $n = 100$, raw PPI and AIPW have MSEs of 0.495 and 0.440, compared with 0.001425 for labeled-only estimation and 0.001391 for `LinearCal`. On PPE Correctness, the corresponding MSEs are 0.534, 0.353, 0.002010, and 0.001813. Relative to labeled-only estimation, `LinearCal` reduces average MSE by 2.4% on PPE Human and 9.8% on PPE Correctness. Its nominal 90% Wald intervals have empirical coverage of 0.909 and 0.931, respectively. Table 1 summarizes the results; evaluator-specific comparisons and model-ranking results appear in Appendices N.3 and N.4.

The ArmoRM results on MATH provide a separate assessment of score ordering and calibration. Empirical correctness increases across twelve score bins, from 0.078 in the lowest bin to 0.920 in the highest. Platt calibration fitted on 100 labels reduces prediction MSE from 0.4683 to 0.2020 on the held-out observations. Appendix N.2 gives the calibration plot and details.

6 Conclusion

Calibrated PPI estimates a population mean by calibrating a prediction score using labeled observations and averaging the calibrated predictions over the pooled covariate sample. Least-squares calibration over a class closed under addition of constants gives an exact AIPW representation. Linear calibration is first-order equivalent to unconstrained PPI++. For isotonic calibration, we establish asymptotic normality and Wald inference without sample splitting, optimality among nondecreasing transformations of the original score, and no further first-order gain from recalibrating the fitted score.

In the LLM benchmarks, linear calibration substantially reduces MSE relative to raw PPI and AIPW, with smaller improvements over labeled-only estimation. In simulations with nonlinear miscalibration, monotone spline and isotonic calibration can further reduce RMSE relative to affine calibration. These comparisons show how the efficiency gains depend on both the original score and the class of calibration maps.

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A LLM evaluation results

For each benchmark, evaluator, and target model, MSE/PPI is the estimator's Monte Carlo MSE divided by that of PPI. Implied label savings are

$$\max\left\{0,\, 1 - \frac{\widehat{\mathrm{Var}}(\widehat{\psi})}{\widehat{\mathrm{Var}}(\overline{Y}_L)}\right\},$$

where the variances are computed over repeated splits and $\overline{Y}_L$ is the labeled mean. We set negative savings to zero before averaging. Table entries are averaged first over target models within each benchmark, then over benchmarks within each evaluator, and then over evaluators.

Appendix Contents

The appendix is ordered from background and practical guidance to extensions, proofs, empirical details, and the implementation sketch.

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Track	n	Estimator	MSE / PPI	Label savings	Coverage
PPE Correctness	100	Labeled-only	0.346	0.000	0.927
PPE Correctness	100	PPI	1.000	0.010	0.904
PPE Correctness	100	AIPW	0.774	0.009	0.906
PPE Correctness	100	LinearCal	0.305	0.101	0.931
PPE Correctness	100	AutoCal	0.310	0.094	0.929
PPE Correctness	400	Labeled-only	0.299	0.000	0.988
PPE Correctness	400	PPI	1.000	0.020	0.924
PPE Correctness	400	AIPW	0.405	0.009	0.941
PPE Correctness	400	LinearCal	0.260	0.105	0.989
PPE Correctness	400	AutoCal	0.263	0.102	0.988
PPE Human Preference	100	Labeled-only	0.303	0.000	0.911
PPE Human Preference	100	PPI	1.000	0.004	0.897
PPE Human Preference	100	AIPW	0.919	0.004	0.896
PPE Human Preference	100	LinearCal	0.295	0.031	0.909
PPE Human Preference	100	AutoCal	0.297	0.028	0.908
PPE Human Preference	400	Labeled-only	0.289	0.000	0.945
PPE Human Preference	400	PPI	1.000	0.005	0.915
PPE Human Preference	400	AIPW	0.703	0.004	0.917
PPE Human Preference	400	LinearCal	0.282	0.028	0.944
PPE Human Preference	400	AutoCal	0.282	0.027	0.944

Table 1: PPE LLM-evaluation summary at $n \in \{100, 400\}$, macro-averaged across the public evaluator models. MSE/PPI reports MSE normalized to raw PPI; implied label savings is relative to labeled-only; coverage is for nominal 90% Wald intervals.

B Extended related work

Although PPI is often presented as a recent development, the underlying estimation problem and many of its estimators are special cases of a broader literature on semiparametric debiasing and regression adjustment, including work on missing data, causal inference, and debiased machine learning.

Semiparametric statistics and missing data. Our problem sits squarely in the missing-data and semiparametric literatures. Rubin [84] formalized missing at random as a key identification condition for recovering full-data targets from partially observed outcomes. Semiparametric efficiency theory then characterized efficient influence functions, regular asymptotically linear estimators, and one-step debiased procedures [54, 69, 36, 48, 8]. In missing-data and causal-inference problems, these ideas led to influence-function-based estimating equations [75, 76, 74, 103, 93], often with doubly robust structure [5]. Accordingly, the efficiency theory for missing-data models under missingness or coarsening at random is well understood, and the class of regular asymptotically linear estimators in these models can be characterized through their influence functions [75, 76, 74, 103, 93].

Debiased machine learning and targeted learning. A large body of work combines the semiparametric framework above with machine-learning-based nuisance estimation, often under the label of debiased machine learning. Flexible learning methods within estimating equations and AIPW-style procedures already appear in van der Laan and Robins [103]. Targeted minimum loss estimation (TMLE), or targeted learning, updates an initial nuisance estimator in a parameter-targeted way so that the resulting plug-in estimator is debiased; in the present setting, the main nuisance is the outcome regression [105, 104, 29, 40, 79]. These approaches are often combined with sample splitting and cross-fitting to accommodate generic machine-learning estimators [88, 48, 116, 104, 18]. Double machine learning emphasizes similar ingredients—orthogonality, cross-fitting, and influence-function expansions—for inference with modern nuisance estimators [18, 20], and can be understood within the same semiparametric framework [102, 23, 16]. For sufficiently stable estimators, cross-fitting can itself be relaxed [13]. Closely related work studies balancing-weight estimators [33, 44, 119, 52, 12, 37], which often admit equivalent regression-adjustment or augmented representations [11, 9, 82]. In completely missing-at-random settings, including semisupervised learning

and randomized trials, these approaches allow flexible black-box nuisance estimation while still supporting valid inference under appropriate conditions [61, 62, 78, 42, 41].

Semisupervised inference. Related work studies semisupervised or surrogate-assisted estimation in settings where the primary outcome is observed only on a limited subset, while auxiliary outcomes or surrogate measurements are available more broadly [67, 68, 15, 14, 17, 45, 47]. Two-sample semisupervised inference is a special case of two-stage sampling designs [90, 77, 38, 72] and, more generally, of data fusion, for which a substantial semiparametric efficiency literature has been developed [55, 56, 28, 112]. More recent papers have used the term *prediction-powered inference* (PPI) for the semisupervised mean-estimation setting in which a black-box predictor is combined with a small labeled sample and a large unlabeled sample [1, 2, 118, 112, 92, 71, 51]. These papers also helped popularize such methods for a broader machine-learning audience and provided accessible software. Closely related questions have long been studied in randomized trials, where prognostic scores or other regression adjustments learned from larger historical or auxiliary data sources are used to improve precision in smaller trials [35, 78, 61–63, 89, 4, 42]. Methodologically, however, PPI estimators are best understood as instances or variants of existing semiparametric procedures. For example, PPI++ is an AIPW estimator [75, 103] obtained by empirical efficiency maximization [83] over a scaling parameter, and cross-PPI [118] is simply AIPW with cross-fitting. MEC [51] instead uses generalized entropy weights to balance the prediction score, which is closer to the balancing-weight representation of our linearly calibrated estimator. Our linearly calibrated estimator emphasizes the corresponding regression-adjustment view, implemented by ordinary linear regression of Y on the score. More fundamentally, the ability to combine black-box prediction models with valid inference is a consequence of the double robustness of AIPW when the missingness probability is known, and is not specific to semisupervised mean estimation [5, 91, 81].

From missing data to causal inference. From the perspective of estimation and efficiency theory, semisupervised mean estimation with missing-at-random outcomes is equivalent to randomized experiments, controlled trials, and survey-sampling designs once the missingness indicator is reinterpreted as a treatment or sampling indicator, so that the relevant potential outcomes are viewed as missing [85, 103, 24, 64]. This equivalence allows tools from covariate adjustment, causal inference, and model-assisted estimation to be transferred directly to the semisupervised setting; see, for example, Mozer [64] for results relating prediction-powered inference to model-assisted survey-sampling estimators. The two-sample formulation we study is also equivalent to a one-sample i.i.d. model [55]. In particular, our methods apply to inference on the counterfactual mean $\mathbb{E}[Y(1)]$ in a randomized trial with binary treatment, where treated units form the labeled sample and control units contribute covariate information only; see Appendix E.

Calibration. A final relevant thread concerns calibration. Classical post-processing methods such as Platt scaling [70, 21, 31], linear calibration [60, 19, 53], isotonic regression [114, 65], and histogram binning [113, 30] improve predictive reliability without retraining the underlying model. The idea has deeper roots in the forecasting literature [60, 57, 109, 108, 27, 49]; see also the historical discussion in Lee et al. [50]. More recent work studies calibration under broader losses and prediction tasks [46, 66, 98, 111, 94], as well as applications to conformal prediction [95, 94], importance-weight stabilization [32, 22, 3, 99], treatment effect estimation [19, 98, 111], and value prediction in offline reinforcement learning [96]. Distribution-free guarantees for histogram binning are developed by Gupta and Ramdas [30], while van der Laan et al. [98], van der Laan and Alaa [94] establish asymptotic distribution-free guarantees for isotonic calibration. Most closely related to our work is recent research connecting calibration to debiased estimation of statistical functionals [100, 97, 101]. In particular, van der Laan et al. [100] studies calibrated debiased machine learning for doubly robust inference, which in our setting reduces to calibrating the outcome regression and exploiting the resulting plug-in representation, while van der Laan et al. [101] studies calibrated plug-in regression estimators in reinforcement learning. More broadly, van der Laan et al. [97] shows how plug-in calibration can support regression and treatment effect estimation, including superefficient inference through dimension reduction.

C Semiparametric background

C.1 Review of AIPW, PPI, and semiparametric efficiency

The AIPW class. We begin with the augmented inverse-probability weighted (AIPW) estimator in the unrestricted two-sample model [75]. We view AIPW not as a single estimator, but as a family of estimators indexed by a regression-adjustment score $q : \mathcal{X} \to \mathbb{R}$ with finite second moment. For any such q, define

$$\widehat{\psi}(q) := \mathbb{P}_M^X q + \mathbb{P}_n^L\{Y - q(X)\}. \tag{10}$$

This estimator combines a plug-in term that imputes Y from $q(X)$ with a residual correction from the labeled sample. For any fixed q, $\widehat{\psi}(q)$ is unbiased in finite samples for $\psi_0 = \mathbb{E}_{P_0}[Y]$. In particular, $q(X) = 0$ yields the empirical mean estimator $\mathbb{P}_n^L\{Y\}$. Thus, the choice of q affects efficiency, through the estimator's variance, but not validity, in the sense that the resulting confidence intervals remain valid. This makes AIPW a "safe" estimator [112]. This property can be viewed as a particularly simple instance of double robustness[2] [5]. The same basic principle underlies the use of flexible machine learning for covariate adjustment in randomized trials while retaining valid inference [61, 62, 89, 42].

Asymptotic linearity and efficiency. We say that $\widehat{\psi}$ is asymptotically linear with triangular influence pair (D_n^L, D_n^U) if

$$\widehat{\psi} - \psi_0 = \frac{1}{n}\sum_{i=1}^{n} D_n^L(X_i, Y_i) + \frac{1}{N}\sum_{j=1}^{N} D_n^U(\widetilde{X}_j) + o_p(n^{-1/2}),$$

where $\mathbb{E}_{P_0}[D_n^L(X,Y)] = 0$ and $\mathbb{E}_{P_{0,X}}[D_n^U(\widetilde{X})] = 0$. If the influence functions converge in L^2 and n/N converges to a finite limit, the central limit theorem and independence of the two samples yield

$$\sqrt{n}\,(\widehat{\psi} - \psi_0) \rightsquigarrow N\Big(0,\ \lim_{n\to\infty}\Big[\mathrm{Var}_{P_0}\{D_n^L(X,Y)\} + \frac{n}{N}\mathrm{Var}_{P_{0,X}}\{D_n^U(\widetilde{X})\}\Big]\Big).$$

The influence pair thus determines the asymptotic variance, accounting for variability in both samples. The following result characterizes the influence pair for the AIPW class, identifies the efficient choice of q, and derives the semiparametric efficiency bound in the unrestricted two-sample model [8, 75, 74].

Proposition 3 (Influence pair and efficiency of AIPW)**.** *Under Assumption 1, for any fixed square-integrable function $q : \mathcal{X} \to \mathbb{R}$, the estimator $\widehat{\psi}(q)$ is asymptotically linear with triangular influence pair $(D_{q,n}^L, D_{q,n}^U)$, where*

$$D_{q,n}^L(X,Y) = Y - \psi_0 - (1-\rho_n)\{q^\circ(X) - \psi_0\},$$
$$D_{q,n}^U(\widetilde{X}) = (1-\rho_n)\{q^\circ(\widetilde{X}) - \psi_0\},$$

and $q^\circ(X) := q(X) - \mathbb{E}_{P_{0,X}}[q(X)] + \psi_0$ is the recentered prediction function. The asymptotic variance of $\sqrt{n}\{\widehat{\psi}(q) - \psi_0\}$ is

$$\rho_0\mathrm{Var}_{P_0}[\mu_0(X)] + \mathbb{E}_{P_0}\big[\mathrm{Var}[Y \mid X]\big] + (1-\rho_0)\mathrm{Var}_{P_0}[q(X) - \mu_0(X)].$$

This variance is minimized when $q = \mu_0$ up to an additive constant. Consequently, the efficient triangular influence pair in the unrestricted two-sample model is $(D_{\mu_0,n}^L, D_{\mu_0,n}^U)$, with efficiency bound $\rho_0\mathrm{Var}_{P_0}[\mu_0(X)] + \mathbb{E}_{P_0}\big[\mathrm{Var}[Y \mid X]\big]$.

By Proposition 3, the limiting variance of the AIPW estimator depends on the score function q only through the mean squared error of its recentered version q°, namely $\mathbb{E}_{P_0}\big[(q^\circ(X) - \mu_0(X))^2\big]$. Thus, within the AIPW class, efficiency is maximized by choosing q so that q° approximates $\mu_0(X)$ as well as possible, with equality when $q^\circ = \mu_0$, in particular at $q = \mu_0$. This shows that full efficiency generally requires using the full covariate information X, or at least a summary rich enough to

[2] In more general missing-at-random settings with unknown, covariate-dependent missingness probabilities, the analogous AIPW estimator remains doubly robust asymptotically, but it no longer enjoys the same finite-sample unbiasedness as in the present known-ρ_n setting [91].

recover $\mu_0(X)$ [74]. In practice, this motivates taking q to be an estimated prediction score $m(X)$, learned using flexible black-box methods.

AIPW, PPI, and empirical-efficiency maximization. The most common choice is $q(X) = m(X)$, where $m(X)$ is a prediction score, yielding the usual augmented inverse-probability weighted (AIPW) estimator [75]:

$$\widehat{\psi}_{\mathrm{AIPW}} = \rho_n \mathbb{P}_n^L\{m(X)\} + (1-\rho_n)\mathbb{P}_N^U\{m(\widetilde{X})\} + \mathbb{P}_n^L\{Y - m(X)\}.$$

Equivalently,

$$\widehat{\psi}_{\mathrm{AIPW}} = \frac{1}{M}\left\{\sum_{i=1}^{n} m(X_i) + \sum_{j=1}^{N} m(\widetilde{X}_j) + \frac{1}{\rho_n}\sum_{i=1}^{n}\{Y_i - m(X_i)\}\right\},$$

where ρ_n^{-1} is the inverse probability of observing the outcome.

Another estimator in this class was considered by Angelopoulos et al. [1] under the name PPI,[3] and we adopt that terminology here. It corresponds to the score choice

$$q(X) = \frac{m(X)}{1-\rho_n}.$$

Substituting this choice into (10) gives

$$\widehat{\psi}_{\mathrm{PPI}} = \mathbb{P}_N^U\{m(\widetilde{X})\} + \mathbb{P}_n^L\{Y - m(X)\}.$$

Relative to $\widehat{\psi}_{\mathrm{AIPW}}$, $\widehat{\psi}_{\mathrm{PPI}}$ uses only the unlabeled sample in the plug-in term and rescales the score by the factor $(1-\rho_n)^{-1} > 1$. By Proposition 3, this helps only if the rescaled score is closer to $\mu_0(X)$ in mean squared error. Thus, PPI is inefficient when $m = \mu_0$, and is typically less efficient when m is already reasonably accurate. When ρ_n is small and m is sufficiently inaccurate, however, neither estimator is necessarily more efficient than the other [34, 80].

Because both $\widehat{\psi}_{\mathrm{AIPW}}$ and $\widehat{\psi}_{\mathrm{PPI}}$ use scores in the scaling class $\{\lambda m(X) : \lambda \in \mathbb{R}\}$, one can choose the scaling factor data-adaptively rather than fixing it in advance. One way to do so is to minimize an empirical estimate of the asymptotic variance of $\sqrt{n}\{\widehat{\psi}(q) - \psi_0\}$:

$$\sigma_n^2(q) := \widehat{\mathrm{Var}}_L\{Y - (1-\rho_n)q(X)\} + \rho_n(1-\rho_n)\,\widehat{\mathrm{Var}}_U\{q(\widetilde{X})\},$$

where $\widehat{\mathrm{Var}}_L(a) := \mathbb{P}_n^L[(a - \mathbb{P}_n^L a)^2]$ and $\widehat{\mathrm{Var}}_U(b) := \mathbb{P}_N^U[(b - \mathbb{P}_N^U b)^2]$. That is, one chooses λ to minimize $\sigma_n^2(q)$ over $q(X) = \lambda m(X)$, a procedure known as empirical-efficiency maximization [83, 103]. The resulting estimator, called PPI++ by Angelopoulos et al. [2], is simply the AIPW estimator with an empirically rescaled prediction rule $\hat{\lambda} m(X)$, where $\hat{\lambda}$ estimates the population-optimal scaling coefficient

$$\lambda^\star = \frac{\mathrm{Cov}\big(Y, m(X)\big)}{\mathrm{Var}\big(m(X)\big)}. \tag{11}$$

The same principle extends beyond simple rescaling. For example, one can minimize $\sigma_n^2(q)$ over richer classes such as $\{\lambda m(X) + \beta_{\mathrm{lin}}^\top X : \lambda \in \mathbb{R},\ \beta_{\mathrm{lin}} \in \mathbb{R}^d\}$, over monotone transformations of the score, or over a reproducing kernel Hilbert space (Appendix D). More generally, the same criterion can be used in cross-validation to choose among candidate estimator classes [83].

Regression adjustment and TMLE can yield the efficient influence pair by updating the regression so that its empirical residual mean vanishes [87, 105, 78]. The calibration identity (4) gives an exact instance of this construction.

D Additional remarks

Remark 1 (Implementation)**.** Isotonic regression is widely available in standard software. A standard algorithm is the pooled-adjacent-violators algorithm (PAVA) [6]. Although it is generally fast, it

[3] The PPI and PPI++ estimators are algebraically equivalent to the model-assisted survey-design estimators of Cassel et al. [10] and Särndal et al. [86], respectively; see Mozer [64] for a review.

may become slow at ultra-large sample sizes. A scalable alternative is to use regression tree software with monotonicity constraints, such as XGBoost or LightGBM (see Appendix O). An advantage of tree-based software is that the fit can be regularized by constraining or tuning the number of leaves, the maximum tree depth, or the minimum number of observations per leaf.

Remark 2 (Platt scaling adjustment)**.** The linear-calibration idea extends beyond squared-error loss. One can instead use logistic loss, Poisson loss, or, more generally, losses from canonical-link generalized linear models [78]. In particular, when $Y \in \{0,1\}$ and $m(X) \in (0,1)$ is an initial predictor, one may apply Platt scaling [21, 70] by fitting a logistic regression of Y on the logit of $m(X)$. Writing $\eta_i(a,b) := a\,\mathrm{logit}(m(X_i)) + b$, this amounts to solving

$$\begin{aligned}\big(\hat{a}_{\mathrm{platt}}, \hat{b}_{\mathrm{platt}}\big) \in \arg\min_{a,b\in\mathbb{R}} \sum_{i=1}^{n} \Big[-Y_i \log \sigma\big(\eta_i(a,b)\big) \\ -(1-Y_i)\log\Big(1-\sigma\big(\eta_i(a,b)\big)\Big)\Big].\end{aligned} \tag{12}$$

with calibrated predictor

$$q_n^{\mathrm{platt}}(X) := \sigma\big(\hat{a}_{\mathrm{platt}}\,\mathrm{logit}(m(X)) + \hat{b}_{\mathrm{platt}}\big),$$

where $\sigma(t) := (1+e^{-t})^{-1}$. Platt scaling may be preferred over linear regression calibration when predicted probabilities are systematically over- or under-confident, in which case the calibration curve is often approximately sigmoidal. More generally, outcomes can be rescaled to $[0,1]$ before applying this procedure. In online settings, calibration properties of Platt scaling are developed in [31].

D.1 Cross-fitting and leave-one-out calibration

Fitting a calibrator to pooled out-of-fold predictions still uses each labeled outcome in the calibration fit. To hold out both steps, partition the labeled indices into folds $I_1, \ldots, I_K$ independently of the observations, and fit each calibrated score $q^{(-k)}$ without the pairs indexed by I_k. Any predictor fitting and tuning used to construct $q^{(-k)}$ also exclude I_k. Let $k(i)$ be the fold containing i and $c_n := N/(n+N)$. The cross-fitted AIPW estimator is

$$\widehat{\psi}_{\mathrm{CF}} := \frac{1}{n}\sum_{i=1}^{n}\left[Y_i + c_n\{\mathbb{P}_N^U q^{(-k(i))} - q^{(-k(i))}(X_i)\}\right]. \tag{13}$$

For each i, conditional on the training sample, $\mathbb{E}[q^{(-k(i))}(X_i)] = \mathbb{E}[\mathbb{P}_N^U q^{(-k(i))}]$. Thus (13) is unbiased when these predictions are integrable and the fitted scores do not use the unlabeled observations.

Leave-one-out calibration takes $K = n$. Writing q_{-i} for the score fitted without (X_i, Y_i), we obtain

$$\widehat{\psi}_{\mathrm{LOO}} = \frac{1}{n}\sum_{i=1}^{n}\left[Y_i + c_n\{\mathbb{P}_N^U q_{-i} - q_{-i}(X_i)\}\right]. \tag{14}$$

The residual correction is retained because the held-out residuals need not average to zero. In the simulation, we use

$$\widehat{\mathrm{se}}^2_{\mathrm{LOO}} = \frac{\widehat{\mathrm{Var}}_L\{Y_i - c_n q_{-i}(X_i)\}}{n} + \frac{\widehat{\mathrm{Var}}_U\{c_n \bar{q}(\widetilde{X}_j)\}}{N}, \qquad \bar{q} := \frac{1}{n}\sum_{i=1}^{n} q_{-i}, \tag{15}$$

where the empirical variances use denominators $n-1$ and $N-1$. The bias, coverage, and interval widths of this procedure are reported in Appendix N.1.

Remark 3 (Empirical efficiency maximization over a prediction class)**.** The empirical variance criterion $\sigma_n^2(q)$ in Section C.1 can be minimized over any candidate class $\mathcal{G} \subset L^2(P_{0,X})$, rather than only over rescalings of one score:

$$\hat{q} \in \arg\min_{q\in\mathcal{G}} \sigma_n^2(q), \qquad \widehat{\psi}_{\mathrm{eem}} := \widehat{\psi}(\hat{q}).$$

At the $\sqrt{n}$ scale, Proposition 3 gives the limiting variance

$$V(q) = \rho_0 \mathrm{Var}[Y] + (1-\rho_0)\mathrm{Var}[Y - q(X)].$$

Writing $q^\circ = q - \mathbb{E}_{P_{0,X}} q + \psi_0$, we have $\mathrm{Var}[Y - q(X)] = \mathbb{E}_{P_0}[(Y - q^\circ(X))^2]$. If $\mathcal{G}$ is closed under constant shifts, population least-squares regression therefore targets a variance-minimizing score. For tuning, average held-out estimates of the influence-function variance across folds and use the selected class or tuning parameter in a cross-fitted AIPW estimator [83].

Remark 4 (Shrinkage via Venn–Abers)**.** When the labeled sample is small, isotonic calibration may be unstable. A natural alternative is Venn–Abers calibration, originally introduced for binary classification [107]; see [95, 94] for generalized versions beyond the binary setting. Assume $Y \in [0, 1]$, or rescale otherwise. For each x, augment the labeled calibration sample with the hypothetical point $(m(x), y)$ for $y \in \{0, 1\}$, fit the resulting isotonic calibrators $f_n^{(x,0)}$ and $f_n^{(x,1)}$, and form the interval-valued prediction

$$\big[f_n^{(x,0)}\{m(x)\},\ f_n^{(x,1)}\{m(x)\}\big].$$

For the Venn–Abers comparator, we shrink the interval midpoint toward the unadjusted AIPW estimator:

$$q_n^{\mathrm{VA}}(x) := q_n^{\mathrm{mid}}(x) + \big\{f_n^{(x,1)}\{m(x)\} - f_n^{(x,0)}\{m(x)\}\big\}\big\{\widehat{\psi}_{AIPW} - q_n^{\mathrm{mid}}(x)\big\},$$

where

$$q_n^{\mathrm{mid}}(x) := \frac{f_n^{(x,1)}\{m(x)\} + f_n^{(x,0)}\{m(x)\}}{2}.$$

The shrinkage weight is the width of the Venn–Abers interval.

Remark 5 (Intercept-only calibration)**.** Set $\hat{a}_{\mathrm{const}} := \mathbb{P}_n^L\{Y - m(X)\}$ and $q_n^{\mathrm{const}} := m + \hat{a}_{\mathrm{const}}$. Then PPI is the unlabeled mean $\mathbb{P}_N^U q_n^{\mathrm{const}}$, whereas AIPW is the pooled mean $\mathbb{P}_M^X q_n^{\mathrm{const}}$.

E Calibrated DML for randomized trials

Let $A \in \{0, 1\}$ denote randomized treatment and $Y = AY(1) + (1 - A)Y(0)$. Assume consistency, no interference, and $A \perp (X, Y(1), Y(0))$, with positive assignment probabilities. When estimating $\mu_a := \mathbb{E}[Y(a)]$, outcomes from arm a are labeled, and covariates from the other arm form the unlabeled sample. Let $\mathbb{P}_{n,a}$ denote the empirical mean in arm a, $\mathbb{P}_n$ the pooled mean, and $\hat{\pi}_a = n_a/n$. If the calibrated arm-specific score satisfies $\mathbb{P}_{n,a}\{Y - m_{n,a}^\star(X)\} = 0$, then

$$\hat{\mu}_a^{\mathrm{cal}} := \mathbb{P}_n\{m_{n,a}^\star(X)\} \tag{16}$$

$$= \mathbb{P}_n\{m_{n,a}^\star(X)\} + \mathbb{P}_n\left[\frac{\mathbf{1}(A = a)}{\hat{\pi}_a}\{Y - m_{n,a}^\star(X)\}\right]. \tag{17}$$

The residual correction is zero because it equals $\mathbb{P}_{n,a}\{Y - m_{n,a}^\star(X)\}$. Thus (4) applies to each arm, and the calibrated treatment-effect estimator is

$$\hat{\tau}^{\mathrm{cal}} := \hat{\mu}_1^{\mathrm{cal}} - \hat{\mu}_0^{\mathrm{cal}}. \tag{18}$$

These arm-specific updates are instances of calibrated DML [100].

F Moment equations under missing at random

For a target satisfying $\mathbb{E}[\phi_{\theta_0}(X, Y)] = 0$, let R indicate an observed outcome, $\pi_0(X) = \Pr(R = 1 \mid X) > 0$, and $a_\theta(X) = \mathbb{E}[\phi_\theta(X, Y) \mid X]$. Under $R \perp Y \mid X$, the augmented moment equation is

$$\mathbb{P}_M\left[a_\theta(X) + \frac{R}{\pi_0(X)}\{\phi_\theta(X, Y) - a_\theta(X)\}\right] = 0.$$

If calibration makes the empirical residual term vanish, this equation reduces to $\mathbb{P}_M a_\theta = 0$. Extensions with estimated missingness probabilities are studied in Robins et al. [75, 76], van der Laan et al. [100].

G Pooled i.i.d. Missing-Data Formulation

When $\rho_0 > 0$, the joint two-sample setup can be embedded into a pooled i.i.d. model by introducing a sample-membership indicator $R \in \{0, 1\}$, where $R = 1$ denotes labeled observations and $R = 0$ denotes unlabeled observations. In that formulation one observes i.i.d. data $O = (X, R, RY)$ with $\mathbb{P}(R = 1) = \rho_0$, and the calibrated estimator becomes

$$\widehat{\psi}_{\text{cal}} = \mathbb{P}_M(q_n) + \mathbb{P}_M\left[\frac{R}{\rho_0}\{Y - q_n\}\right]$$

up to the same calibration identity as in the main text. The efficient influence function in the reduced model observing only the score is

$$\mathbb{E}[Y \mid m(X)] - \psi_0 + \frac{R}{\rho_0}\{Y - \mathbb{E}[Y \mid m(X)]\},$$

which is the single-sample analogue of the efficient influence pair obtained by applying proposition 3 to the reduced model with scalar covariate $m(X)$. Thus the usual i.i.d. missing-data formulation is a convenient special case, but the main text keeps the two-dataset structure explicit.

H AIPW comparison identity

Lemma 1 (AIPW comparison identity)**.** *Suppose* $\mathbb{P}_n^L\{Y - q_n(X)\} = 0$*. For any comparison score* g *with finite empirical averages,*

$$\widehat{\psi}_{\text{cal}} = \widehat{\psi}(g) + (1 - \rho_n)(\mathbb{P}_N^U - \mathbb{P}_n^L)(q_n - g).$$

Proof. The residual equation gives $\widehat{\psi}_{\text{cal}} = \widehat{\psi}(q_n)$. Subtracting $\widehat{\psi}(g)$ and collecting terms proves the identity. □

I PPI++ and linear calibration

Proof of Proposition 2. Write

$$\hat{v}_L := \mathbb{P}_n^L[(m - \mathbb{P}_n^L m)^2], \quad \hat{v}_U := \mathbb{P}_N^U[(m - \mathbb{P}_N^U m)^2], \quad \hat{c}_L := \mathbb{P}_n^L[(Y - \mathbb{P}_n^L Y)(m - \mathbb{P}_n^L m)].$$

On the event $\hat{v}_L > 0$, the least-squares normal equations give

$$\hat{a}_{\text{lin}} = \frac{\hat{c}_L}{\hat{v}_L}, \qquad \hat{b}_{\text{lin}} = \mathbb{P}_n^L Y - \hat{a}_{\text{lin}}\mathbb{P}_n^L m.$$

Differentiating the empirical variance criterion in Section C.1 gives

$$\hat{\lambda}_{++} = \frac{\hat{c}_L}{(1 - \rho_n)\hat{v}_L + \rho_n \hat{v}_U}.$$

Finite second moments and $\text{Var}[m(X)] > 0$ imply $\hat{v}_L, \hat{v}_U \to_p \text{Var}[m(X)]$ and $\hat{c}_L \to_p \text{Cov}(Y, m(X))$. Thus $\Pr(\hat{v}_L > 0) \to 1$, both coefficients converge to $a_0 := \text{Cov}(Y, m(X))/\text{Var}[m(X)]$, and $\hat{\lambda}_{++} - \hat{a}_{\text{lin}} = o_p(1)$.

For $\Delta_n := \mathbb{P}_N^U m - \mathbb{P}_n^L m$, substituting the fitted intercept gives

$$\widehat{\psi}_{\text{lin}} = \mathbb{P}_n^L Y + (1 - \rho_n)\hat{a}_{\text{lin}}\Delta_n, \qquad \widehat{\psi}_{++} = \mathbb{P}_n^L Y + (1 - \rho_n)\hat{\lambda}_{++}\Delta_n.$$

Subtracting proves the exact identity in the proposition. Independence and finite second moments give $\Delta_n = O_p(n^{-1/2} + N^{-1/2})$. Since $\rho_n \to \rho_0 < 1$, the ratio n/N is bounded, so

$$\widehat{\psi}_{++} - \widehat{\psi}_{\text{lin}} = (1 - \rho_n)o_p(1)\Delta_n = o_p(n^{-1/2}).$$

□

J Proof of Proposition 3

Proof of Proposition 3. Adding a constant to q leaves $\widehat{\psi}(q)$ unchanged. Thus $\widehat{\psi}(q) = \widehat{\psi}(q^\circ)$, where $q^\circ = q - \mathbb{E}_{P_{0,X}} q + \psi_0$. Centering the empirical means gives the exact identity

$$\widehat{\psi}(q) - \psi_0 = (\mathbb{P}_n^L - P_0)D_{q,n}^L + (\mathbb{P}_N^U - P_{0,X})D_{q,n}^U.$$

Both components have mean zero. Write $A = \mu_0 - \psi_0$, $\delta = q^\circ - \mu_0$, and $\varepsilon = Y - \mu_0$. Then

$$D_{q,n}^L = \varepsilon + \rho_n A - (1-\rho_n)\delta, \qquad D_{q,n}^U = (1-\rho_n)(A+\delta).$$

Since $\mathbb{E}[\varepsilon \mid X] = 0$ and $n/N = \rho_n/(1-\rho_n)$, the covariance terms cancel in

$$\mathrm{Var}(D_{q,n}^L) + \frac{n}{N}\mathrm{Var}(D_{q,n}^U) = \mathbb{E}[\mathrm{Var}(Y \mid X)] + \rho_n \mathrm{Var}(\mu_0) + (1-\rho_n)\mathrm{Var}(\delta).$$

Taking limits gives the stated variance, since $\mathrm{Var}(\delta) = \mathrm{Var}(q - \mu_0)$. The choice $q = \mu_0$ removes the last term and gives the missing-data efficiency bound [76, 74, 93]. □

K Supporting lemmas for calibration

Lemma 2 (Blockwise residual orthogonality)**.** *Let $B_1, \ldots, B_J$ be the pooled adjacent violator blocks of the isotonic fit, and let $q_n^{\mathrm{iso}}(X_i) = c_j$ for $i \in B_j$. Then*

$$\sum_{i \in B_j} (Y_i - c_j) = 0, \qquad j = 1, \ldots, J.$$

Consequently, for every measurable function $h : \mathbb{R} \to \mathbb{R}$ such that $h(q_n^{\mathrm{iso}}(X_i))$ is finite for all i,

$$\sum_{i=1}^n h(q_n^{\mathrm{iso}}(X_i))\{Y_i - q_n^{\mathrm{iso}}(X_i)\} = 0.$$

Proof. The fitted values from isotonic regression are constant on pooled adjacent violator blocks. The KKT conditions imply that residuals sum to zero on each block. Since $h(q_n^{\mathrm{iso}}(X_i)) = h(c_j)$ for all $i \in B_j$,

$$\sum_{i=1}^n h(q_n^{\mathrm{iso}}(X_i))\{Y_i - q_n^{\mathrm{iso}}(X_i)\} = \sum_{j=1}^J h(c_j) \sum_{i \in B_j} (Y_i - c_j) = 0.$$

□

Proof of proposition 1. Lemma 2 gives (5). By definition, the fitted isotonic score minimizes the empirical squared loss among all nondecreasing transformations of m, proving the risk comparison. Taking the identity transformation gives the comparison with the raw score. □

Lemma 3 (Bounded variation after isotonic post-processing)**.** *Suppose Assumption 3 holds. Let $T := m(X)$ and $S_n := q_n^{\mathrm{iso}}(X)$. Define*

$$\eta(t) := \mathbb{E}_{P_0}[Y \mid T = t], \qquad \eta_n(s) := \mathbb{E}_{P_0}[Y \mid S_n = s].$$

If $t \mapsto \eta(t)$ has total variation at most V, then, conditional on the labeled sample, the map $s \mapsto \eta_n(s)$ has total variation at most V. If, in addition, $|\eta(t)| \le C_0$ on the support of T, then $|\eta_n(s)| \le C_0$ on the support of S_n.

Proof. We adapt the proof of Lemma 6 in [98]. Let $S_n = f_n(T)$, where f_n is the fitted isotonic map. Conditional on the labeled sample, f_n is nondecreasing. Hence, for each s in the support of S_n, there exists an interval $B_s \subset \mathbb{R}$, possibly a singleton, such that

$$\{S_n = s\} = \{T \in B_s\}.$$

Therefore

$$\eta_n(s) = \mathbb{E}_{P_0}[Y \mid S_n = s] = \mathbb{E}_{P_0}[\eta(T) \mid T \in B_s].$$

Let $\eta = \eta^+ - \eta^-$ be the Jordan decomposition of η, where η^+ and η^- are nondecreasing and

$$\|\eta\|_{TV} = \|\eta^+\|_{TV} + \|\eta^-\|_{TV} \le V.$$

Then

$$\eta_n(s) = \mathbb{E}_{P_0}[\eta^+(T) \mid T \in B_s] - \mathbb{E}_{P_0}[\eta^-(T) \mid T \in B_s].$$

If $s_1 < s_2$, then monotonicity of f_n implies every point of B_{s_1} is no larger than every point of B_{s_2}. Hence the conditional distribution of $T \mid T \in B_{s_2}$ first-order stochastically dominates that of $T \mid T \in B_{s_1}$. Therefore, for any nondecreasing function g,

$$\mathbb{E}_{P_0}[g(T) \mid T \in B_{s_1}] \le \mathbb{E}_{P_0}[g(T) \mid T \in B_{s_2}].$$

Applying this with $g = \eta^+$ and $g = \eta^-$, both maps

$$s \mapsto \mathbb{E}_{P_0}[\eta^+(T) \mid T \in B_s] \qquad \text{and} \qquad s \mapsto \mathbb{E}_{P_0}[\eta^-(T) \mid T \in B_s]$$

are nondecreasing.

Moreover, each of these conditional expectations takes values in the range of the corresponding function, so

$$\big\|s \mapsto \mathbb{E}_{P_0}[\eta^+(T) \mid T \in B_s]\big\|_{TV} \le \|\eta^+\|_{TV}, \qquad \big\|s \mapsto \mathbb{E}_{P_0}[\eta^-(T) \mid T \in B_s]\big\|_{TV} \le \|\eta^-\|_{TV}.$$

Hence

$$\|\eta_n\|_{TV} \le \|\eta^+\|_{TV} + \|\eta^-\|_{TV} = \|\eta\|_{TV} \le V.$$

Finally, if $|\eta(t)| \le C_0$ on the support of T, then

$$|\eta_n(s)| = |\mathbb{E}_{P_0}[\eta(T) \mid T \in B_s]| \le C_0$$

for all s in the support of S_n. □

Lemma 4 (The population isotonic projection is centered)**.** *Under Assumptions 1 and 2,* $\mathbb{E}_{P_{0,X}}[q_0^{\mathrm{iso}}(X)] = \psi_0$.

Proof. The isotonic class is closed under constant shifts, so $c \mapsto \mathbb{E}[(Y - q_0^{\mathrm{iso}}(X) - c)^2]$ is minimized at $c = 0$. Its derivative there is $-2\mathbb{E}[Y - q_0^{\mathrm{iso}}(X)]$, giving the stated identity. □

Proof convention. In the following proofs, fix $\zeta > 0$. By $\sup_x |q_n^{\mathrm{iso}}(x)| = O_p(1)$, there exist a deterministic constant $C = C_\zeta \ge C_0$ and $n_0 < \infty$ such that the event

$$A_{n,C} := \Big\{\sup_x |q_n^{\mathrm{iso}}(x)| \le C\Big\}$$

satisfies $P_0(A_{n,C}) \ge 1 - \zeta$ for all $n \ge n_0$. We carry out the deterministic bounded-class arguments on $A_{n,C}$, where all implicit constants may depend on C but not on n. Since ζ is arbitrary, the resulting rates and distributional statements hold unconditionally.

Lemma 5 (Isotonic L^2 rate)**.** *Suppose Assumptions 1 and 2 holds. Then*

$$\|q_n^{\mathrm{iso}} - q_0^{\mathrm{iso}}\|_{2,P_{0,X}}^2 = O_p(n^{-2/3}).$$

Proof of Lemma 5. Let $T := m(X)$, $\eta(T) := \mathbb{E}_{P_0}[Y \mid T]$, and

$$\mathcal{M}_C := \{f(T) : f \in \mathcal{F}_{\mathrm{iso}},\ \|f(T)\|_\infty \le C\}.$$

By Assumption 2, $\eta(T)$ is bounded by $C_0 \le C$, and on $A_{n,C}$ we have $q_n^{\mathrm{iso}} \in \mathcal{M}_C$. Since clipping any monotone candidate to $[-C, C]$ preserves monotonicity and can only decrease squared risk against $\eta(T)$, the population isotonic projection q_0^{iso} also belongs to $\mathcal{M}_C$. For every $m \in \mathcal{M}_C$,

$$\mathbb{E}_{P_0}[(Y - m)^2] = \mathbb{E}_{P_0}[(Y - \eta)^2] + \mathbb{E}_{P_0}[(\eta - m)^2],$$

which shows that q_0^{iso} is the $L^2(P_T)$-projection of η onto the closed convex set $\mathcal{M}_C$.

Now define the centered class

$$\mathcal{F}_0 := \{g = m - q_0^{\mathrm{iso}} :\ m \in \mathcal{M}_C\}.$$

For $g \in \mathcal{F}_0$, let $\ell_g(Z) := (Y - q_0^{\mathrm{iso}}(X) - g(X))^2 - (Y - q_0^{\mathrm{iso}}(X))^2$, where $Z = (X, Y)$. Since q_n^{iso} minimizes the empirical squared risk over $\mathcal{F}_{\mathrm{iso}}$, on $A_{n,C}$ the random element $\hat{g} := q_n^{\mathrm{iso}} - q_0^{\mathrm{iso}}$ belongs to $\mathcal{F}_0$ and satisfies

$$\mathbb{P}_n^L\{\ell_{\hat{g}}\} \le 0 = \mathbb{P}_n^L\{\ell_0\}.$$

Moreover,

$$\mathbb{E}_{P_0}[\ell_g] = \|g\|_{2,P_{0,X}}^2 - 2\mathbb{E}_{P_0}[(Y - q_0^{\mathrm{iso}})g].$$

Because g is measurable with respect to T, the tower property gives

$$\mathbb{E}_{P_0}[(Y - q_0^{\mathrm{iso}})g] = \mathbb{E}_{P_0}[(\eta - q_0^{\mathrm{iso}})g].$$

Since q_0^{iso} is the $L^2(P_T)$-projection of η onto the convex set $\mathcal{M}_C$, the Hilbert-space projection inequality yields

$$\mathbb{E}_{P_0}[(\eta - q_0^{\mathrm{iso}})(m - q_0^{\mathrm{iso}})] \le 0 \qquad \text{for every } m \in \mathcal{M}_C.$$

Hence

$$\mathbb{E}_{P_0}[\ell_g] \ge \|g\|_{2,P_{0,X}}^2,$$

so squared loss has the required quadratic margin around q_0^{iso}.

We next bound the modulus of continuity of the empirical process indexed by the localized classes

$$\mathcal{F}_\delta := \{g \in \mathcal{F}_0 : \|g\|_{2,P_{0,X}} \le \delta\}.$$

The bracketing bound for bounded monotone functions gives

$$\log N_{[]}(\varepsilon, \mathcal{F}_\delta, L_2(P_{0,X})) \lesssim \varepsilon^{-1}, \qquad 0 < \varepsilon < \delta.$$

Consequently,

$$J_{[]}(\delta, \mathcal{F}_\delta, L_2(P_{0,X})) \lesssim \delta^{1/2};$$

see, e.g., van der Vaart and Wellner [106].

Write $\xi := Y - q_0^{\mathrm{iso}}(X)$, so that $\ell_g = g^2 - 2\xi g$. By Condition (ii) and the bounds on η, q_0^{iso}, and g, there exists $A_C < \infty$ such that

$$\mathbb{E}_{P_0}[|\xi g|^k] \le k! A_C^k \|g\|_{2,P_{0,X}}^2, \qquad g \in \mathcal{F}_0, \quad k \ge 2. \tag{19}$$

The same inequality holds for bounded bracket widths. Thus multiplication by ξ maps $L^2(P_{0,X})$ brackets to Bernstein-norm brackets with widths bounded by a constant multiple of their original widths; see Lemmas 8–9 of Bibaut and van der Laan [7]. The squared class $\{g^2 : g \in \mathcal{F}_\delta\}$ is a bounded Lipschitz image of $\mathcal{F}_\delta$. Applying Lemmas 3.4.2–3.4.3 of van der Vaart and Wellner [106] to these two classes gives

$$E^\star \sup_{g \in \mathcal{F}_\delta} |(\mathbb{P}_n^L - P_0)\ell_g| \lesssim n^{-1/2}\delta^{1/2} + n^{-1}\delta^{-1}. \tag{20}$$

For $\delta \ge n^{-1/3}$, the right-hand side is bounded by a constant times $n^{-1/2}\delta^{1/2}$.

Set $\varepsilon_n := n^{-1/3}$. The empirical risk inequality, the quadratic margin, and (20) imply, by peeling into shells $2^s\varepsilon_n \le \|\hat{g}\|_{2,P_{0,X}} < 2^{s+1}\varepsilon_n$ and applying Markov's inequality, that

$$\begin{aligned}\Pr\{\|\hat{g}\|_{2,P_{0,X}} \ge 2^S\varepsilon_n,\ A_{n,C}\} &\lesssim \sum_{s \ge S} \frac{n^{-1/2}(2^{s+1}\varepsilon_n)^{1/2}}{2^{2s}\varepsilon_n^2} \\ &\lesssim \sum_{s \ge S} 2^{-3s/2} \longrightarrow 0 \qquad (S \to \infty).\end{aligned}$$

Hence

$$\|q_n^{\mathrm{iso}} - q_0^{\mathrm{iso}}\|_{2,P_{0,X}} = O_p(n^{-1/3}), \qquad \|q_n^{\mathrm{iso}} - q_0^{\mathrm{iso}}\|_{2,P_{0,X}}^2 = O_p(n^{-2/3}).$$

Since $P_0(A_{n,C}^c) \le \zeta$ for all large n, and $\zeta > 0$ was arbitrary, the same rate holds unconditionally, proving Lemma 5. □

Theorem 3 (Calibration error of isotonic post-processing)**.** *Suppose Assumptions 1 to 3 hold. Let*

$$q_{0|n}(X) := \mathbb{E}_{P_0}[Y \mid q_n^{\mathrm{iso}}(X)].$$

Then

$$\|q_{0|n} - q_n^{\mathrm{iso}}\|_{2,P_{0,X}}^2 = O_p(n^{-2/3}).$$

Proof. The proof follows that of Theorem 1 in [98]; see also [94]. Let

$$T := m(X), \qquad S_n := q_n^{\mathrm{iso}}(X), \qquad \eta_n(s) := \mathbb{E}_{P_0}[Y \mid S_n = s],$$

and define

$$\Delta_n(X) := q_{0|n}(X) - q_n^{\mathrm{iso}}(X) = \eta_n(S_n) - S_n.$$

Conditional on the labeled sample, $h_n(s) := \eta_n(s) - s$ is a measurable function of s. Applying proposition 1 pathwise with $h = h_n$ yields

$$\mathbb{P}_n^L\{\Delta_n(X)\{Y - S_n\}\} = 0.$$

Also,

$$\mathbb{E}_{P_0}[\Delta_n(X)\{Y - S_n\}] = \mathbb{E}_{P_0}[\Delta_n(X)\,\mathbb{E}_{P_0}[Y - S_n \mid S_n]] = \mathbb{E}_{P_0}[\Delta_n^2] = \|\Delta_n\|_{2,P_{0,X}}^2.$$

Hence

$$\|\Delta_n\|_{2,P_{0,X}}^2 = (\mathbb{E}_{P_0} - \mathbb{P}_n^L)\{\Delta_n(X)\{Y - S_n\}\}.$$

Fix $\zeta > 0$, and work on the event $A_{n,C}$ from the proof convention. On this event, S_n is uniformly bounded by C. By Lemma 3, the function η_n belongs to the bounded-variation class

$$\mathcal{F}_{\mathrm{TV},C} := \{f : \mathbb{R} \to \mathbb{R} : \|f\|_\infty \le C,\ \|f\|_{TV} \le V\},$$

where $V < \infty$ is the variation bound in Assumption 3. Let

$$\mathcal{F}_{\mathrm{iso},C} := \{f : \mathbb{R} \to \mathbb{R} : f \text{ nondecreasing and } \|f\|_\infty \le C\}.$$

Since $S_n = f_n(T)$ for a nondecreasing fitted isotonic map f_n, and composition with a monotone map preserves bounded variation, the function

$$t \mapsto \eta_n\{f_n(t)\}$$

also belongs to $\mathcal{F}_{\mathrm{TV},C}$, after enlarging C if necessary. Thus $\eta_n(S_n) = h_1(X)$ for some $h_1 = f_1 \circ m$ with $f_1 \in \mathcal{F}_{\mathrm{TV},C}$, while $S_n = h_2(X)$ for $h_2 = f_2 \circ m$ with $f_2 = f_n \in \mathcal{F}_{\mathrm{iso},C}$.

Define the localized product class

$$\mathcal{G}_{n,C,\delta} := \Big\{(h_1 - h_2)(Y - h_2) : h_1 = f_1 \circ m,\ h_2 = f_2 \circ m,\\ f_1 \in \mathcal{F}_{\mathrm{TV},C},\ f_2 \in \mathcal{F}_{\mathrm{iso},C},\ \|h_1 - h_2\|_{2,P_{0,X}} \le \delta\Big\}.$$

Then

$$\Delta_n(X)\{Y - S_n\} \in \mathcal{G}_{n,C,\delta_n}, \qquad \delta_n := \|\Delta_n\|_{2,P_{0,X}}.$$

Write $\eta(T) := \mathbb{E}_{P_0}[Y \mid T]$ and $\varepsilon := Y - \eta(T)$. Each element of $\mathcal{G}_{n,C,\delta}$ decomposes as

$$(h_1 - h_2)(Y - h_2) = (h_1 - h_2)\varepsilon + (h_1 - h_2)\{\eta(T) - h_2\}.$$

Bounded-variation functions are differences of bounded monotone functions, so their bracketing entropy is $O(\epsilon^{-1})$. The second term is a bounded Lipschitz transformation of (h_1, h_2). For the first term, Condition (ii) gives the moment bound in (19), with $h_1 - h_2$ in place of g and ε in place of ξ. Its Bernstein-norm bracket widths are therefore controlled by $L^2(P_{0,X})$ bracket widths. The second term has $L^2(P_0)$ norm $O(\delta)$, and the first has Bernstein norm $O(\delta)$; both bracketing entropy integrals are $O(\delta^{1/2})$. Lemmas 3.4.2–3.4.3 of van der Vaart and Wellner [106] give

$$E^\star \sup_{g \in \mathcal{G}_{n,C,\delta}} |(\mathbb{P}_n^L - P_0)\{g\}| \lesssim n^{-1/2}\delta^{1/2} + n^{-1}\delta^{-1}.$$

For $\delta \ge n^{-1/3}$, this is bounded by a constant times $n^{-1/2}\delta^{1/2}$.

Set $\varepsilon_n := n^{-1/3}$. For $s \ge 0$, let

$$B_s := \{2^s\varepsilon_n \le \delta_n < 2^{s+1}\varepsilon_n\}.$$

On $B_s \cap A_{n,C}$,

$$2^{2s}\varepsilon_n^2 \le \delta_n^2 \le \sup_{g \in \mathcal{G}_{n,C,2^{s+1}\varepsilon_n}} |(\mathbb{P}_n^L - P_0)\{g\}|.$$

Therefore, by Markov's inequality,

$$P_0(B_s \cap A_{n,C}) \lesssim \frac{E^\star \sup_{g \in \mathcal{G}_{n,C,2^{s+1}\varepsilon_n}} |(\mathbb{P}_n^L - P_0)\{g\}|}{2^{2s}\varepsilon_n^2} \lesssim \frac{n^{-1/2}(2^{s+1}\varepsilon_n)^{1/2}}{2^{2s}\varepsilon_n^2} \lesssim 2^{-3s/2},$$

since $n^{-1/2}\varepsilon_n^{-3/2} = 1$. Summing over $s \geq S$ yields

$$P_0(\delta_n \geq 2^S \varepsilon_n,\ A_{n,C}) \lesssim \sum_{s=S}^{\infty} 2^{-3s/2} \xrightarrow[S\to\infty]{} 0.$$

Because $P_0(A_{n,C}^c) \leq \zeta$ for all large n, and $\zeta > 0$ is arbitrary, it follows that

$$\|\Delta_n\|_{2,P_{0,X}} = O_p(n^{-1/3}), \qquad \|\Delta_n\|_{2,P_{0,X}}^2 = O_p(n^{-2/3}).$$

This is exactly the claimed calibration-error rate. □

Lemma 6 (Centered empirical-process bounds for isotonic calibration)**.** *Suppose Assumptions 1 and 2 holds. Then*

$$(\mathbb{P}_n^L - P_{0,X})\{q_n^{\text{iso}}(X) - q_0^{\text{iso}}(X)\} = O_p(n^{-2/3}),$$
$$(\mathbb{P}_N^U - P_{0,X})\{q_n^{\text{iso}}(\widetilde{X}) - q_0^{\text{iso}}(\widetilde{X})\} = O_p\big(n^{-1/3}N^{-1/2}\big).$$

Proof of Lemma 6. By Lemma 5,

$$\|q_n^{\text{iso}} - q_0^{\text{iso}}\|_{2,P_{0,X}}^2 = O_p(n^{-2/3}).$$

To control the centered empirical-process term, work on the event $A_{n,C}$ from the proof convention. For fixed $R < \infty$, consider the localized class

$$\mathcal{F}_{n,C} := \{g = f \circ m - q_0^{\text{iso}} :\ f \in \mathcal{F}_{\text{iso}},\ \|f \circ m\|_\infty \leq C,\ \|g\|_{2,P_{0,X}} \leq Rn^{-1/3}\}.$$

By Lemma 5, the probability of $A_{n,C} \cap \{q_n^{\text{iso}} - q_0^{\text{iso}} \notin \mathcal{F}_{n,C}\}$ can be made arbitrarily small by taking R large. Because m is fixed, $\mathcal{F}_{n,C}$ is a translated class of bounded monotone functions of a one-dimensional score. The monotone bracketing bound implies

$$J_{[]}(r, \mathcal{F}_{n,C}, L_2(P_0)) \lesssim r^{1/2}.$$

Applying Lemma 3.4.2 of van der Vaart and Wellner [106] at radius $r_n = Rn^{-1/3}$ yields

$$\sup_{g \in \mathcal{F}_{n,C}} |(\mathbb{P}_n^L - P_{0,X})\{g\}| = O_p\Big(n^{-1/2}r_n^{1/2} + n^{-1}r_n^{-1}\Big) = O_p(n^{-2/3}).$$

In particular,

$$(\mathbb{P}_n^L - P_{0,X})\{q_n^{\text{iso}} - q_0^{\text{iso}}\} = O_p(n^{-2/3}).$$

Since $P_0(A_{n,C}^c) \leq \zeta$ for all large n, and $\zeta > 0$ is arbitrary, the same bound holds unconditionally.

For the unlabeled sample, conditional on the labeled data, $q_n^{\text{iso}} - q_0^{\text{iso}}$ is fixed and the unlabeled sample is independent, so

$$(\mathbb{P}_N^U - P_{0,X})\{q_n^{\text{iso}} - q_0^{\text{iso}}\} = O_p\left(N^{-1/2}\|q_n^{\text{iso}} - q_0^{\text{iso}}\|_{2,P_{0,X}}\right) = O_p\big(n^{-1/3}N^{-1/2}\big)$$

by Lemma 5. □

Lemma 7 (Centered empirical-process bounds for isotonic post-processing)**.** *Suppose Assumptions 1 to 3 hold. Let*

$$\Delta_n(X) := q_{0|n}(X) - q_n^{\text{iso}}(X).$$

Then

$$\big(\mathbb{P}_n^L - P_{0,X}\big)\{\Delta_n(X)\} = O_p(n^{-2/3}), \qquad \big(\mathbb{P}_N^U - P_{0,X}\big)\{\Delta_n(\widetilde{X})\} = O_p\big(n^{-1/3}N^{-1/2}\big).$$

Proof. By theorem 3,

$$\|\Delta_n\|_{2,P_{0,X}}^2 = O_p(n^{-2/3}).$$

Let $V < \infty$ be the variation bound in Assumption 3. Fix $\zeta > 0$, and work on the event $A_{n,C}$. On this event, q_n^{iso} is uniformly bounded by C. By Lemma 3, the function

$$s \mapsto \eta_n(s) := \mathbb{E}_{P_0}[Y \mid q_n^{\text{iso}}(X) = s]$$

has total variation at most V and is bounded by C on the support of $q_n^{\text{iso}}(X)$. Since $q_n^{\text{iso}}(X) = f_n(T)$ with $T = m(X)$ and f_n nondecreasing, it follows that

$$q_{0|n}(X) = \eta_n\{f_n(T)\}$$

is a bounded-variation transform of T, while $q_n^{\text{iso}}(X)$ is a bounded monotone transform of T.

Consider the classes

$$\mathcal{F}_{\text{TV},C} := \{f : \mathbb{R} \to \mathbb{R} : \|f\|_\infty \le C,\ \|f\|_{TV} \le V\},$$
$$\mathcal{F}_{\text{iso},C} := \{f : \mathbb{R} \to \mathbb{R} : f \text{ nondecreasing and } \|f\|_\infty \le C\}.$$

For fixed $R < \infty$, define

$$\mathcal{H}_{n,C} := \Big\{h_1 - h_2 : h_1 = f_1 \circ m,\ h_2 = f_2 \circ m,\ f_1 \in \mathcal{F}_{\text{TV},C},\ f_2 \in \mathcal{F}_{\text{iso},C},\ \|h_1 - h_2\|_{2,P_{0,X}} \le Rn^{-1/3}\Big\}.$$

By theorem 3, the probability of $A_{n,C} \cap \{\Delta_n \notin \mathcal{H}_{n,C}\}$ can be made arbitrarily small by taking R large.

Because bounded-variation functions are differences of bounded monotone functions, the localized bracketing entropy of $\mathcal{H}_{n,C}$ obeys the same bound as in the proof of theorem 3, namely

$$J_{[]}(\delta, \mathcal{H}_{n,C}, L_2(P_0)) \lesssim \delta^{1/2}.$$

Applying Lemma 3.4.2 of van der Vaart and Wellner [106] at radius $r_n = Rn^{-1/3}$ gives

$$\sup_{h \in \mathcal{H}_{n,C}} |(\mathbb{P}_n^L - P_{0,X})h| = O_p\Big(n^{-1/2} r_n^{1/2} + n^{-1} r_n^{-1}\Big) = O_p(n^{-2/3}).$$

Taking R large gives the same bound for $(\mathbb{P}_n^L - P_{0,X})\Delta_n$ on $A_{n,C}$. Since $P_0(A_{n,C}^c) \le \zeta$ for all large n, and $\zeta > 0$ was arbitrary, the same bound holds unconditionally.

For the unlabeled sample, conditional on the labeled data, Δ_n is fixed and the unlabeled sample is independent. Hence

$$\big(\mathbb{P}_N^U - P_{0,X}\big)\{\Delta_n(\widetilde{X})\} = O_p\left(N^{-1/2}\|\Delta_n\|_{2,P_{0,X}}\right) = O_p\big(n^{-1/3}N^{-1/2}\big)$$

by theorem 3. □

L Proof of asymptotic linearity

Proof of theorem 1. By Lemma 4, $\mathbb{E}_{P_{0,X}}[q_0^{\text{iso}}] = \psi_0$. The isotonic fit has mean-zero labeled residuals by Lemma 2. Applying Lemma 1 with $g = q_0^{\text{iso}}$ gives

$$\widehat{\psi}_{\text{iso}} = \rho_n \mathbb{P}_n^L\left\{q_0^{\text{iso}} + \rho_n^{-1}(Y - q_0^{\text{iso}})\right\} + (1-\rho_n)\mathbb{P}_N^U\{q_0^{\text{iso}}(\widetilde{X})\} + (1-\rho_n)(\mathbb{P}_N^U - \mathbb{P}_n^L)\{q_n^{\text{iso}} - q_0^{\text{iso}}\}.$$

Subtracting ψ_0 gives

$$\begin{aligned}\widehat{\psi}_{\text{iso}} - \psi_0 = (\mathbb{P}_n^L - P_0)D^L_{q_0^{\text{iso}},n} + (\mathbb{P}_N^U - P_{0,X})D^U_{q_0^{\text{iso}},n} \\ + (1-\rho_n)(\mathbb{P}_N^U - \mathbb{P}_n^L)\{q_n^{\text{iso}} - q_0^{\text{iso}}\}.\end{aligned}$$

Also,

$$\begin{aligned}(1-\rho_n)(\mathbb{P}_N^U - \mathbb{P}_n^L)\{q_n^{\text{iso}} - q_0^{\text{iso}}\} = (1-\rho_n)(\mathbb{P}_N^U - P_{0,X})\{q_n^{\text{iso}} - q_0^{\text{iso}}\} \\ - (1-\rho_n)(\mathbb{P}_n^L - P_{0,X})\{q_n^{\text{iso}} - q_0^{\text{iso}}\}.\end{aligned}$$

By Lemma 6, the labeled term is $O_p(n^{-2/3})$ and the unlabeled term is $O_p(n^{-1/3}N^{-1/2})$. Hence the remainder satisfies the bound in equation (9). Moreover,

$$\sqrt{n}(1-\rho_n)n^{-2/3} \le n^{-1/6}$$

and, writing $t_n = N/n$,

$$\sqrt{n}(1-\rho_n)n^{-1/3}N^{-1/2} = n^{-1/3}\frac{\sqrt{t_n}}{1+t_n} \le \frac{1}{2}n^{-1/3}.$$

Therefore $R_{n,N} = o_p(n^{-1/2})$, which proves equation (9). □

Proof of corollary 1. By theorem 1,

$$\sqrt{n}\,(\widehat{\psi}_{\text{iso}} - \psi_0) = \frac{1}{\sqrt{n}}\sum_{i=1}^{n} D^L_{q_0^{\text{iso}},n}(X_i, Y_i) + \frac{\sqrt{n}}{N}\sum_{j=1}^{N} D^U_{q_0^{\text{iso}},n}(\widetilde{X}_j) + o_p(1).$$

The labeled and unlabeled samples are independent and each summand has mean zero. Moreover,

$$\frac{n}{N}\text{Var}\{D^U_{q_0^{\text{iso}},n}(\widetilde{X})\} = \rho_n(1-\rho_n)\text{Var}\{q_0^{\text{iso}}(X)\}.$$

Since $\rho_n \to \rho_0 \in [0,1)$, a triangular-array central limit theorem yields the stated asymptotic normal law with variance σ_0^2.

For the variance estimator, let $g_n := q_n^{\text{iso}} - q_0^{\text{iso}}$. By Lemma 5,

$$\mathbb{E}_{P_{0,X}}[g_n^2] = \|g_n\|^2_{2,P_{0,X}} = O_p(n^{-2/3}).$$

Applying Lemma 3.4.2 of van der Vaart and Wellner [106] to the squared class gives

$$\sup_{g\in\mathcal{F}_\delta} |(\mathbb{P}_n^L - P_{0,X})\{g^2\}| = O_p\Big(n^{-1/2}\delta^{1/2} + n^{-1}\delta^{-1}\Big),$$

where $\mathcal{F}_\delta$ is the localized class from the proof of Lemma 5. By that lemma, on $A_{n,C}$ the function g_n belongs to $\mathcal{F}_{Rn^{-1/3}}$ with arbitrarily high probability for sufficiently large fixed R. Taking $\delta = Rn^{-1/3}$ gives

$$(\mathbb{P}_n^L - P_{0,X})\{g_n^2\} = O_p(n^{-2/3}).$$

Hence

$$\mathbb{P}_n^L(g_n^2) = O_p(n^{-2/3}) = o_p(1).$$

On the boundedness event $A_{n,C}$, conditional on the labeled data, g_n is fixed and $|g_n| \le 2C$, so

$$(\mathbb{P}_N^U - P_{0,X})\{g_n^2\} = O_p\left(N^{-1/2}\|g_n^2\|_{2,P_{0,X}}\right) = O_p\left(C\,N^{-1/2}\|g_n\|_{2,P_{0,X}}\right) = O_p\big(n^{-1/3}N^{-1/2}\big).$$

Therefore

$$\mathbb{P}_N^U(g_n^2) = O_p(n^{-2/3}) + O_p\big(n^{-1/3}N^{-1/2}\big) = o_p(1).$$

Since $P_0(A_{n,C}^c) \le \zeta$ for all large n and arbitrary $\zeta > 0$, these bounds hold unconditionally. Also, $\widehat{\psi}_{\text{iso}} \to_p \psi_0$ follows from theorem 1.

Finally,

$$\widehat{D}^L - D^L_{q_0^{\text{iso}},n} = -(1-\rho_n)g_n - \rho_n(\widehat{\psi}_{\text{iso}} - \psi_0),$$

and

$$\widehat{D}^U - D^U_{q_0^{\text{iso}},n} = (1-\rho_n)\{g_n(\widetilde{X}) - (\widehat{\psi}_{\text{iso}} - \psi_0)\}.$$

Because $\mathbb{P}_n^L(g_n^2), \mathbb{P}_N^U(g_n^2) = o_p(1)$ and $\widehat{\psi}_{\text{iso}} \to_p \psi_0$, it follows that

$$\mathbb{P}_n^L\Big[\{\widehat{D}^L - D^L_{q_0^{\text{iso}},n}\}^2\Big] = o_p(1), \qquad \frac{n}{N}\mathbb{P}_N^U\Big[\{\widehat{D}^U - D^U_{q_0^{\text{iso}},n}\}^2\Big] = o_p(1),$$

where the second conclusion uses

$$\frac{n}{N}(1-\rho_n)^2 = \rho_n(1-\rho_n) \le \frac{1}{4}.$$

The law of large numbers and $\rho_n \to \rho_0$ also give

$$\mathbb{P}_n^L\{(D^L_{q_0^{\text{iso}},n})^2\} + \frac{n}{N}\mathbb{P}_N^U\{(D^U_{q_0^{\text{iso}},n})^2\} \;\to_p\; \sigma_0^2.$$

The empirical means of both estimated influence components converge to zero, so within-sample centering does not change this limit. Therefore $\widehat{\sigma}^2 \to_p \sigma_0^2$. The Wald interval claim then follows by Slutsky's theorem. □

M Proof of post-processing equivalence and reduced-model efficiency

Proof of Theorem 2. Set $\Delta_n = q_{0|n} - q_n^{\mathrm{iso}}$. Lemma 1 with $g = q_{0|n}$ gives

$$\widehat{\psi}_{\mathrm{eff}} - \widehat{\psi}_{\mathrm{iso}} = (1-\rho_n)(\mathbb{P}_N^U - \mathbb{P}_n^L)\Delta_n.$$

By Lemma 7, the right-hand side is

$$O_p((1-\rho_n)n^{-2/3}) + O_p((1-\rho_n)n^{-1/3}N^{-1/2}) = o_p(n^{-1/2}),$$

using the same bounds as in the proof of Theorem 1. The calibration-error rate is Theorem 3. Together these prove both conclusions. □

Proof of corollary 2. For part (i), if $q_0^{\mathrm{iso}} = \mu_0$ almost surely, then the asymptotic influence pair in theorem 1 is

$$D^L_{\mu_0,n}(X,Y) = Y - \psi_0 - (1-\rho_n)\{\mu_0(X) - \psi_0\}, \qquad D^U_{\mu_0,n}(\widetilde{X}) = (1-\rho_n)\{\mu_0(\widetilde{X}) - \psi_0\}.$$

By proposition 3, this is the efficient influence pair in the unrestricted two-sample model. Therefore $\widehat{\psi}_{\mathrm{iso}}$ is efficient for ψ_0 in the full two-sample experiment.

For part (ii), fix a square-integrable nondecreasing transformation $g = \theta \circ m$, and let $g^\circ = g - \mathbb{E}_{P_{0,X}} g + \psi_0$. Adding a constant preserves monotonicity, so population isotonic optimality and Lemma 4 give

$$\mathrm{Var}(Y - q_0^{\mathrm{iso}}) = \mathbb{E}[(Y - q_0^{\mathrm{iso}})^2] \le \mathbb{E}[(Y - g^\circ)^2] = \mathrm{Var}(Y - g).$$

Proposition 3 now gives the efficiency comparison.

For part (iii), let

$$S := m(X), \qquad \eta_0(s) := \mathbb{E}_{P_0}[Y \mid S = s].$$

proposition 3 applied to the reduced two-sample experiment with scalar covariate S shows that the efficient triangular influence pair in that reduced experiment is

$$D^L_{\eta_0,n}(S,Y) = Y - \psi_0 - (1-\rho_n)\{\eta_0(S) - \psi_0\}, \qquad D^U_{\eta_0,n}(\widetilde{S}) = (1-\rho_n)\{\eta_0(\widetilde{S}) - \psi_0\}.$$

If η_0 is monotone increasing, then η_0 belongs to the isotonic class and

$$\mathbb{E}_{P_0}\left[\{Y - g(S)\}^2\right] = \mathbb{E}_{P_0}\left[\{Y - \eta_0(S)\}^2\right] + \mathbb{E}_{P_0}\left[\{\eta_0(S) - g(S)\}^2\right]$$

for every measurable g. Hence the population isotonic regression of Y on the score $S = m(X)$ is exactly $\eta_0(S)$, so

$$q_0^{\mathrm{iso}}(X) = \eta_0\{m(X)\} \qquad \text{almost surely.}$$

Therefore, the influence pair in theorem 1 coincides with the reduced-model efficient influence pair above, so $\widehat{\psi}_{\mathrm{iso}}$ is efficient for ψ_0 in the reduced experiment indexed by $m(X)$. The isotonic class is closed under constant shifts and positive rescaling. The corresponding population first-order conditions give $\mathbb{E}[Y - q_0^{\mathrm{iso}}] = \mathbb{E}[q_0^{\mathrm{iso}}(Y - q_0^{\mathrm{iso}})] = 0$, so $\mathrm{Cov}(Y, q_0^{\mathrm{iso}}) = \mathrm{Var}(q_0^{\mathrm{iso}})$. Substitution into σ_0^2 gives the stated variance reduction relative to the labeled mean. □

N Additional empirical details

N.1 Small-sample comparisons and RePPI

The small-sample experiment uses $S \sim N(0,1)$, $p(S) = \{1 + \exp(-5S)\}^{-1}$, and $Y \mid S \sim \mathrm{Bernoulli}\{p(S)\}$. Let $C(t) = \min\{0.99, \max(0.01, t)\}$, $\sigma(t) = (1+e^{-t})^{-1}$, and $Z \sim N(0,1)$ independently of (S,Y). The mildly miscalibrated score is $C[\sigma\{1.5\,\mathrm{logit}(C[p(S) + 0.005Z])\}]$; the strongly miscalibrated score is $C[-0.15 + 0.75\,\sigma\{3\,\mathrm{logit}(p(S)) - 0.5\}]$. The simulation seed is 20260411. Figure 3 reports signed bias, coverage of nominal 95% intervals, interval width, and relative efficiency versus PPI++.

We implement two scalar-score versions of RePPI Algorithm 1 [45], using three-way sample splitting and weights proportional to the evaluation-fold sizes. RePPI-RF uses a 100-tree random forest. RePPI-MLP-CV selects hidden-layer widths from (8), $(32,32)$, and $(32,32,32)$, and penalties from

$\{10^{-2}, 10^{-4}\}$, by two-fold validation MSE within each outer training fold. Both estimate the scalar power-tuning coefficient on the evaluation fold, use the global factor $N/(n+N)$, and set the coefficient to zero when fitted-score variance is numerically singular. The neural implementation uses the feature vector $[1, m(X)]$. Its prediction map and the scalar single-fold calculation were checked against the public implementation; the three-way aggregation follows the paper's Algorithm 1.

RePPI-RF has lower RMSE than PPI++ in one of 14 settings; RePPI-MLP-CV does so in five. For $n \geq 50$, the neural estimator's RMSE is 0.576–0.668 times that of the labeled mean. At $n = 20$, the ratios are 1.450 and 9.690 under mild and strong miscalibration. Neither RePPI implementation has lower RMSE than smooth monotone calibration in these settings. The largest neural-estimator error occurs when a fitted-score variance of 1.04×10^{-7} produces a coefficient of 1238.2. All replications are retained. Singular-variance fallbacks occur in 175 random-forest replications and 46 neural-network replications; the neural fits produce 2,120 convergence warnings.

This appendix describes the benchmark construction used in section 5. We compare PPI with AIPW, clipped PPI++ with AIPW-EM, and `LinearCal` with `AutoCal`.

We use the official `ppi_py` real-data mean-estimation examples and preserve their labeled-sample-size grids. The `forest` experiment uses $n \in \{50, 100, \ldots, 500\}$. The `galaxies` and semisupervised `census_income` benchmarks use $n \in \{50, 155, 261, 366, 472, 577, 683, 788, 894, 1000\}$. For each n, the remaining observations form the unlabeled sample. We rerandomize this split 500 times at each n, always evaluating all estimators on the same split and taking the full-sample mean as the benchmark truth.

We compare the labeled-only mean, PPI, AIPW, PPI++, AIPW-EM, `LinearCal`, `MonoSpline`, `IsoCal`, and `AutoCal`. The binary-outcome benchmarks also include imputation, Platt scaling, and Venn–Abers shrinkage. PPI and PPI++ use the official `ppi_py` implementations. PPI++ restricts its coefficient to $[0, 1]$, whereas AIPW-EM minimizes the empirical variance without that constraint.

Calibration maps are fitted on the labeled pairs. We use the calibrated scores in the AIPW estimator, retaining the residual correction after clipping or binning. `LinearCal` fits an affine map; `MonoSpline` fits a smooth nondecreasing spline with a roughness penalty. Both clip predictions to the range of the labeled outcomes. `IsoCal` uses a minimum bin size of 10. Platt scaling fits logistic regression to a stabilized logit transform of the score. `AutoCal` selects among AIPW, linear calibration, monotone splines, and isotonic calibration using 20-fold cross-validation of the estimated influence-function variance. Each fold uses an unlabeled subsample of size $\min(N, 10n)$ to evaluate this criterion. We report Monte Carlo bias, variance, MSE, Wald-interval coverage, and relative efficiency versus PPI.

N.2 LLM score ordering and held-out calibration

For MATH responses evaluated by ArmoRM, empirical correctness increases across twelve score bins, from 0.078 in the lowest bin to 0.920 in the highest. Platt calibration fitted on 100 randomly selected labels reduces mean squared prediction error from 0.4683 to 0.2020 on the remaining held-out observations, a reduction of 56.9%.

N.3 LLM evaluator breakdown

The largest raw-score MSE inflation occurs for Skywork and Athene. The other evaluator summaries show smaller differences among the main-text estimators. The evaluator-specific results show where calibration provides the largest gains.

N.4 LLM ranking supplement

The PPE human-preference track also induces a downstream ranking problem across the selected target models. Using the same repeated labeled/unlabeled splits as in section 5, we therefore summarize how well each estimator recovers the target-model ordering induced by the full-data human win shares. For each evaluator, labeled-sample size, replicate, and estimator, we rank the eight selected target models by estimated win share, compare that ordering against the ranking induced by the full-data targets, and then macro-average across evaluators. We report Spearman rank correlation, top-1 identification rate, and top-1 regret, defined as the difference in true win share between the best

Algorithm 1 Isotonic-calibrated PPI

Require: $\mathcal{D}_L = \{(X_i, Y_i)\}_{i=1}^n$, $\mathcal{D}_U = \{\widetilde{X}_j\}_{j=1}^N$, score m, level $1-\alpha$

1: Compute $T_i = m(X_i)$ and $\widetilde{T}_j = m(\widetilde{X}_j)$
2: Fit $\hat{f} \in \arg\min_{f \in \mathcal{F}_{\text{iso}}} \sum_{i=1}^n \{Y_i - f(T_i)\}^2$
3: Set $q_n^{\text{iso}}(X_i) = \hat{f}(T_i)$ and $q_n^{\text{iso}}(\widetilde{X}_j) = \hat{f}(\widetilde{T}_j)$
4: Form $\widehat{\psi}_{\text{iso}} \leftarrow (n+N)^{-1}\{\sum_{i=1}^n q_n^{\text{iso}}(X_i) + \sum_{j=1}^N q_n^{\text{iso}}(\widetilde{X}_j)\}$
5: Set $\widehat{D}_i^L = Y_i - \widehat{\psi}_{\text{iso}} - (1-\rho_n)\{q_n^{\text{iso}}(X_i) - \widehat{\psi}_{\text{iso}}\}$ and $\widehat{D}_j^U = (1-\rho_n)\{q_n^{\text{iso}}(\widetilde{X}_j) - \widehat{\psi}_{\text{iso}}\}$
6: Set $\overline{D}_L = n^{-1}\sum_{i=1}^n \widehat{D}_i^L$ and $\overline{D}_U = N^{-1}\sum_{j=1}^N \widehat{D}_j^U$
7: Set $\widehat{\text{SE}}_{\text{iso}}^2 = n^{-2}\sum_{i=1}^n (\widehat{D}_i^L - \overline{D}_L)^2 + N^{-2}\sum_{j=1}^N (\widehat{D}_j^U - \overline{D}_U)^2$
8: Return $\widehat{\psi}_{\text{iso}} \pm z_{1-\alpha/2}\widehat{\text{SE}}_{\text{iso}}$

target model and the model selected by the estimator. The main pattern mirrors the mean-estimation results: calibrated and efficiency-maximized estimators substantially outperform raw PPI/AIPW, while remaining close to labeled-only at small n. At $n = 200$, `AutoCal` attains the highest average Spearman correlation, about 0.905; at $n = 400$, `LinearCal` gives the best overall ranking quality, with Spearman about 0.941 and top-1 identification about 0.646.

N.5 Data provenance, compute, and reproducibility

All empirical results use public benchmark sources cited in the main text and appendix. The tabular reproductions use the official `ppi_py` mean-estimation datasets; the LLM experiments use PPE Human Preference V1 and PPE correctness data derived from MMLU-Pro, MATH, GPQA, IFEval, and MBPP+. The submission does not introduce a new dataset or foundation model. Figures and tables are generated from the experiment scripts and cached CSV summaries; raw public data and evaluator-score sources remain governed by their upstream licenses and terms.

The main simulation uses 500 Monte Carlo repetitions per setting, and the small-sample comparison uses 1,000. The reproduced PPI benchmarks use 500 random labeled/unlabeled splits for each combination of dataset and labeled sample size. The PPE experiments use 200 random splits for each combination of evaluator, benchmark, and target model. We use public precomputed evaluator scores.

The local runs used Python 3.9.6 with `numpy` 2.0.2, `pandas` 2.3.3, `scikit-learn` 1.6.1, `xgboost` 2.1.4, `lightgbm` 4.6.0, `matplotlib` 3.9.4, and `ppi-python` 0.2.3. Computations used an Apple M1 Pro machine with 10 logical CPUs and 16 GB RAM. The 1,000-repetition small-sample experiment used eight parallel workers and took 9,214 seconds of elapsed time.

O Implementation sketch

Reference implementations are in `experiments/` in the supplementary code. The algorithms below use the notation of the paper.

O.1 Algorithmic summaries

Algorithm 2 Linearly calibrated PPI

Require: $\mathcal{D}_L = \{(X_i, Y_i)\}_{i=1}^n$, $\mathcal{D}_U = \{\widetilde{X}_j\}_{j=1}^N$, score m, level $1-\alpha$

1: Compute $T_i = m(X_i)$ and $\widetilde{T}_j = m(\widetilde{X}_j)$
2: Fit $(\hat{a}, \hat{b}) \in \arg\min_{a,b} \sum_{i=1}^n \{Y_i - (aT_i + b)\}^2$
3: Set $q_n^{\text{lin}}(X_i) = \hat{a}T_i + \hat{b}$ and $q_n^{\text{lin}}(\widetilde{X}_j) = \hat{a}\widetilde{T}_j + \hat{b}$
4: Form $\widehat{\psi}_{\text{lin}} \leftarrow (n+N)^{-1}\{\sum_{i=1}^n q_n^{\text{lin}}(X_i) + \sum_{j=1}^N q_n^{\text{lin}}(\widetilde{X}_j)\}$
5: Set $\widehat{D}_i^L = Y_i - \widehat{\psi}_{\text{lin}} - (1-\rho_n)\{q_n^{\text{lin}}(X_i) - \widehat{\psi}_{\text{lin}}\}$ and $\widehat{D}_j^U = (1-\rho_n)\{q_n^{\text{lin}}(\widetilde{X}_j) - \widehat{\psi}_{\text{lin}}\}$
6: Set $\overline{D}_L = n^{-1}\sum_{i=1}^n \widehat{D}_i^L$ and $\overline{D}_U = N^{-1}\sum_{j=1}^N \widehat{D}_j^U$
7: Set $\widehat{\text{SE}}_{\text{lin}}^2 = n^{-2}\sum_{i=1}^n (\widehat{D}_i^L - \overline{D}_L)^2 + N^{-2}\sum_{j=1}^N (\widehat{D}_j^U - \overline{D}_U)^2$
8: Return $\widehat{\psi}_{\text{lin}} \pm z_{1-\alpha/2}\widehat{\text{SE}}_{\text{lin}}$

Algorithm 3 Linearly calibrated PPI with covariate adjustment

Require: $\mathcal{D}_L = \{(X_i, Y_i)\}_{i=1}^n$, $\mathcal{D}_U = \{\widetilde{X}_j\}_{j=1}^N$, score m, level $1-\alpha$

1: Compute $T_i = m(X_i)$ and $\widetilde{T}_j = m(\widetilde{X}_j)$
2: Fit $(\hat{a}, \hat{\beta}, \hat{b}) \in \arg\min_{a,\beta,b} \sum_{i=1}^n \{Y_i - (aT_i + X_i^\top \beta + b)\}^2$
3: Set $q_n^{\text{lin}}(X_i) = \hat{a}T_i + X_i^\top\hat{\beta} + \hat{b}$ and $q_n^{\text{lin}}(\widetilde{X}_j) = \hat{a}\widetilde{T}_j + \widetilde{X}_j^\top\hat{\beta} + \hat{b}$
4: Form $\widehat{\psi}_{\text{lin}} \leftarrow (n+N)^{-1}\{\sum_{i=1}^n q_n^{\text{lin}}(X_i) + \sum_{j=1}^N q_n^{\text{lin}}(\widetilde{X}_j)\}$
5: Set $\widehat{D}_i^L = Y_i - \widehat{\psi}_{\text{lin}} - (1-\rho_n)\{q_n^{\text{lin}}(X_i) - \widehat{\psi}_{\text{lin}}\}$ and $\widehat{D}_j^U = (1-\rho_n)\{q_n^{\text{lin}}(\widetilde{X}_j) - \widehat{\psi}_{\text{lin}}\}$
6: Set $\overline{D}_L = n^{-1}\sum_{i=1}^n \widehat{D}_i^L$ and $\overline{D}_U = N^{-1}\sum_{j=1}^N \widehat{D}_j^U$
7: Set $\widehat{\text{SE}}_{\text{lin}}^2 = n^{-2}\sum_{i=1}^n (\widehat{D}_i^L - \overline{D}_L)^2 + N^{-2}\sum_{j=1}^N (\widehat{D}_j^U - \overline{D}_U)^2$
8: Return $\widehat{\psi}_{\text{lin}} \pm z_{1-\alpha/2}\widehat{\text{SE}}_{\text{lin}}$

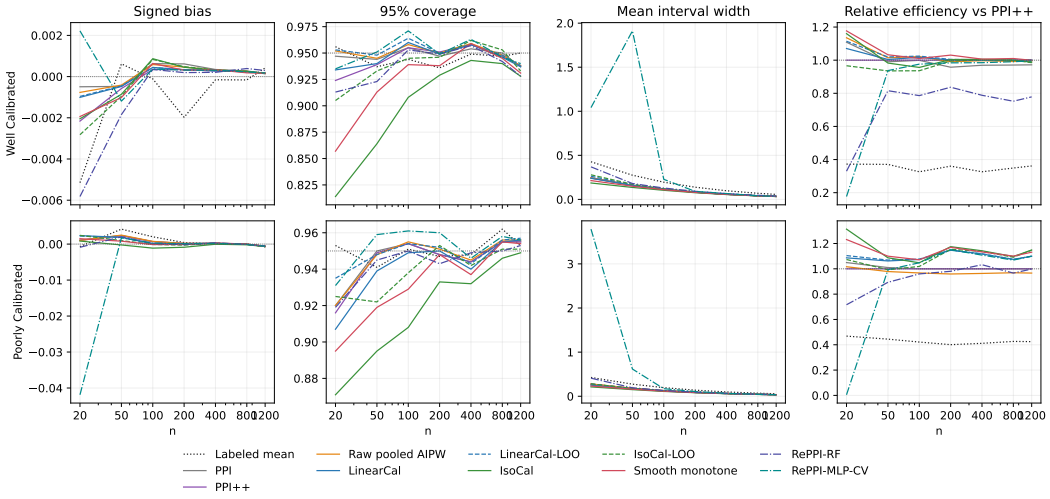


Figure 3: Signed bias, coverage of nominal 95% intervals, interval width, and relative efficiency versus PPI++ in the small-sample experiment. The numerical summary includes Monte Carlo standard errors.

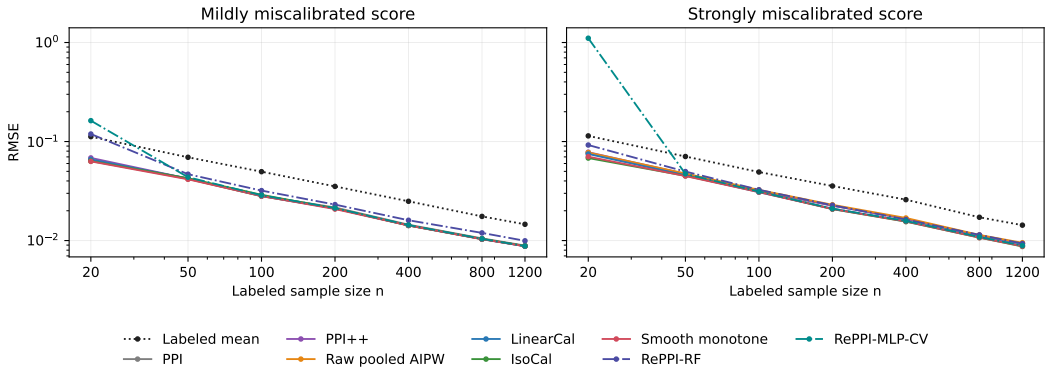


Figure 4: Small-sample RMSE comparison including the random-forest and cross-validated neural-network RePPI implementations.

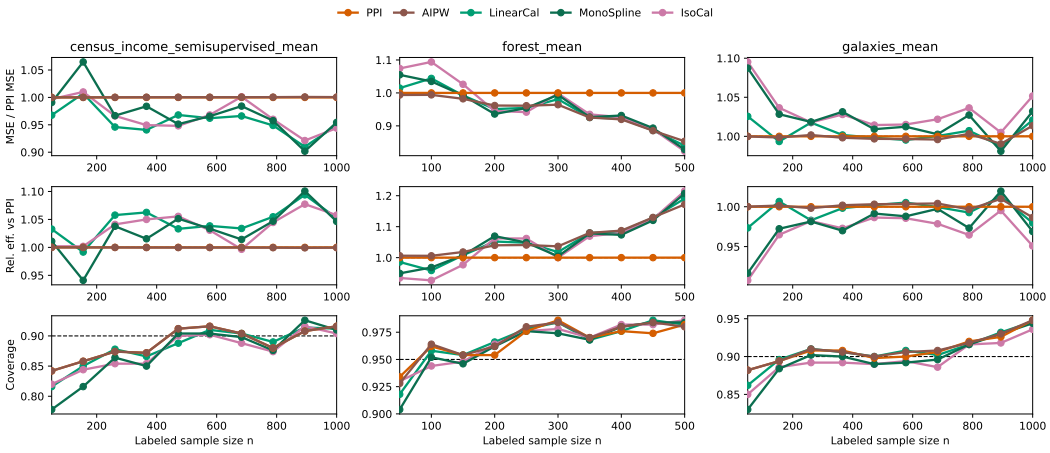


Figure 5: Calibration-focused appendix comparison across the reproduced PPI benchmarks, restricted to PPI, AIPW, `LinearCal`, `MonoSpline`, and `IsoCal`. The panels report normalized MSE relative to PPI, relative efficiency versus PPI, and coverage across labeled-sample-size regimes.

Dataset	n	N	Estimator	Bias	Variance	MSE	Coverage	RelEff
census_income_semisupervised_mean	50	380041	AIPW	356.4905	69836665.4410	69963750.9366	0.842	1.000
census_income_semisupervised_mean	50	380041	AIPW-EM	260.9833	67728426.6042	67796538.8913	0.822	1.031
census_income_semisupervised_mean	50	380041	AutoCal	-123.2501	69242944.1124	69258134.7078	0.832	1.009
census_income_semisupervised_mean	50	380041	Labeled-only	685.3532	105558402.2427	106028111.3137	0.860	0.662
census_income_semisupervised_mean	50	380041	IsoCal	-215.3832	69697400.2434	69743790.1536	0.820	1.002
census_income_semisupervised_mean	50	380041	LinearCal	260.9294	67622601.4006	67690685.5430	0.816	1.033
census_income_semisupervised_mean	50	380041	MonoSpline	-482.0477	69082916.1270	69315286.1508	0.778	1.011
census_income_semisupervised_mean	50	380041	PPI	356.4473	69834865.9486	69961920.5977	0.842	1.000
census_income_semisupervised_mean	50	380041	PPI++	150.8664	72953053.8328	72975814.4980	0.830	0.957
census_income_semisupervised_mean	577	379514	AIPW	199.8136	6353884.5833	6393810.0648	0.916	1.000
census_income_semisupervised_mean	577	379514	AIPW-EM	177.6067	6114918.2974	6146462.4536	0.912	1.039
census_income_semisupervised_mean	577	379514	AutoCal	134.3746	6201029.6459	6219086.1828	0.910	1.024
census_income_semisupervised_mean	577	379514	Labeled-only	213.2465	9407595.6062	9453069.6946	0.902	0.675
census_income_semisupervised_mean	577	379514	IsoCal	155.0503	6162964.7883	6187005.3977	0.902	1.031
census_income_semisupervised_mean	577	379514	LinearCal	179.8005	6116832.9552	6149161.1594	0.910	1.038
census_income_semisupervised_mean	577	379514	MonoSpline	162.1952	6146069.6864	6172376.9541	0.904	1.033
census_income_semisupervised_mean	577	379514	PPI	199.7932	6351836.6166	6391753.9369	0.916	1.000
census_income_semisupervised_mean	577	379514	PPI++	198.7409	6362047.6345	6401545.5819	0.916	0.998
census_income_semisupervised_mean	1000	379091	AIPW	92.1091	3462051.4638	3470535.5425	0.916	0.999
census_income_semisupervised_mean	1000	379091	AIPW-EM	61.4705	3288235.4312	3292014.0588	0.912	1.052
census_income_semisupervised_mean	1000	379091	AutoCal	25.8207	3289467.9320	3290134.6381	0.922	1.052
census_income_semisupervised_mean	1000	379091	Labeled-only	165.4763	5293296.2822	5320678.6872	0.914	0.654
census_income_semisupervised_mean	1000	379091	IsoCal	51.2697	3271622.4413	3274251.0223	0.904	1.058
census_income_semisupervised_mean	1000	379091	LinearCal	60.7501	3287796.5296	3291487.1058	0.912	1.052
census_income_semisupervised_mean	1000	379091	MonoSpline	60.9626	3306575.6859	3310292.1289	0.910	1.046
census_income_semisupervised_mean	1000	379091	PPI	91.9155	3460071.7259	3468520.1894	0.916	1.000
census_income_semisupervised_mean	1000	379091	PPI++	91.9155	3460071.7259	3468520.1894	0.916	1.000
forest_mean	50	1546	AIPW	-0.0012	0.0018	0.0018	0.928	1.006
forest_mean	50	1546	AIPW-EM	-0.0018	0.0019	0.0019	0.918	0.969
forest_mean	50	1546	AutoCal	-0.0028	0.0019	0.0019	0.924	0.965
forest_mean	50	1546	Labeled-only	0.0017	0.0024	0.0024	0.954	0.762
forest_mean	50	1546	Imputation	-0.0733	0.0000	0.0054	0.000	–
forest_mean	50	1546	IsoCal	-0.0030	0.0020	0.0020	0.930	0.934
forest_mean	50	1546	LinearCal	-0.0016	0.0019	0.0019	0.918	0.986
forest_mean	50	1546	MonoSpline	-0.0021	0.0019	0.0019	0.904	0.949
forest_mean	50	1546	Platt	-0.0003	0.0019	0.0019	0.900	0.948
forest_mean	50	1546	PPI	-0.0013	0.0018	0.0018	0.934	1.000
forest_mean	50	1546	PPI++	-0.0064	0.0019	0.0019	0.928	0.969
forest_mean	50	1546	Venn-Abers	-0.0027	0.0020	0.0020	0.888	0.923
forest_mean	300	1296	AIPW	0.0011	0.0002	0.0002	0.984	1.036
forest_mean	300	1296	AIPW-EM	0.0007	0.0002	0.0002	0.982	1.012
forest_mean	300	1296	AutoCal	0.0006	0.0002	0.0002	0.978	1.002
forest_mean	300	1296	Labeled-only	0.0007	0.0003	0.0004	0.968	0.706
forest_mean	300	1296	Imputation	-0.0733	0.0000	0.0054	0.000	–
forest_mean	300	1296	IsoCal	0.0010	0.0002	0.0002	0.978	1.000
forest_mean	300	1296	LinearCal	0.0010	0.0002	0.0002	0.984	1.018
forest_mean	300	1296	MonoSpline	0.0010	0.0002	0.0002	0.974	1.004
forest_mean	300	1296	Platt	0.0010	0.0002	0.0002	0.982	1.020
forest_mean	300	1296	PPI	0.0011	0.0002	0.0002	0.986	1.000
forest_mean	300	1296	PPI++	-0.0002	0.0002	0.0002	0.980	0.991
forest_mean	300	1296	Venn-Abers	0.0011	0.0002	0.0002	0.978	1.015
forest_mean	500	1096	AIPW	0.0001	0.0001	0.0001	0.980	1.172
forest_mean	500	1096	AIPW-EM	-0.0002	0.0001	0.0001	0.980	1.188
forest_mean	500	1096	AutoCal	-0.0003	0.0001	0.0001	0.986	1.180
forest_mean	500	1096	Labeled-only	0.0001	0.0002	0.0002	0.972	0.879
forest_mean	500	1096	Imputation	-0.0733	0.0000	0.0054	0.000	–
forest_mean	500	1096	IsoCal	-0.0000	0.0001	0.0001	0.986	1.215
forest_mean	500	1096	LinearCal	0.0001	0.0001	0.0001	0.982	1.191
forest_mean	500	1096	MonoSpline	0.0000	0.0001	0.0001	0.984	1.207
forest_mean	500	1096	Platt	0.0001	0.0001	0.0001	0.984	1.203
forest_mean	500	1096	PPI	0.0000	0.0002	0.0002	0.982	1.000
forest_mean	500	1096	PPI++	-0.0006	0.0001	0.0001	0.978	1.179
forest_mean	500	1096	Venn-Abers	-0.0001	0.0001	0.0001	0.978	1.206
galaxies_mean	50	16693	AIPW	0.0035	0.0018	0.0018	0.882	1.000
galaxies_mean	50	16693	AIPW-EM	0.0035	0.0019	0.0020	0.864	0.944
galaxies_mean	50	16693	AutoCal	0.0019	0.0018	0.0018	0.896	0.997
galaxies_mean	50	16693	Labeled-only	0.0014	0.0041	0.0041	0.890	0.451
galaxies_mean	50	16693	Imputation	-0.0433	0.0000	0.0019	0.000	–
galaxies_mean	50	16693	IsoCal	-0.0011	0.0020	0.0020	0.850	0.908
galaxies_mean	50	16693	LinearCal	0.0030	0.0019	0.0019	0.862	0.973
galaxies_mean	50	16693	MonoSpline	0.0026	0.0020	0.0020	0.830	0.917
galaxies_mean	50	16693	Platt	0.0033	0.0019	0.0019	0.856	0.966
galaxies_mean	50	16693	PPI	0.0035	0.0018	0.0018	0.882	1.000
galaxies_mean	50	16693	PPI++	-0.0020	0.0020	0.0020	0.864	0.914
galaxies_mean	50	16693	Venn-Abers	0.0022	0.0020	0.0020	0.830	0.907
galaxies_mean	577	16166	AIPW	-0.0006	0.0001	0.0001	0.906	1.004
galaxies_mean	577	16166	AIPW-EM	-0.0007	0.0001	0.0001	0.906	1.005
galaxies_mean	577	16166	AutoCal	-0.0008	0.0001	0.0001	0.918	0.989
galaxies_mean	577	16166	Labeled-only	-0.0011	0.0003	0.0003	0.910	0.436
galaxies_mean	577	16166	Imputation	-0.0433	0.0000	0.0019	0.000	–
galaxies_mean	577	16166	IsoCal	-0.0007	0.0001	0.0001	0.894	0.986
galaxies_mean	577	16166	LinearCal	-0.0007	0.0001	0.0001	0.908	1.005
galaxies_mean	577	16166	MonoSpline	-0.0006	0.0001	0.0001	0.892	0.988
galaxies_mean	577	16166	Platt	-0.0007	0.0001	0.0001	0.896	1.002
galaxies_mean	577	16166	PPI	-0.0006	0.0001	0.0001	0.900	1.000
galaxies_mean	577	16166	PPI++	-0.0012	0.0001	0.0001	0.902	0.988
galaxies_mean	577	16166	Venn-Abers	-0.0006	0.0001	0.0001	0.892	0.986
galaxies_mean	1000	15743	AIPW	0.0002	0.0001	0.0001	0.948	0.987
galaxies_mean	1000	15743	AIPW-EM	0.0002	0.0001	0.0001	0.948	0.980
galaxies_mean	1000	15743	AutoCal	0.0001	0.0001	0.0001	0.944	0.977
galaxies_mean	1000	15743	Labeled-only	0.0003	0.0002	0.0002	0.884	0.357
galaxies_mean	1000	15743	Imputation	-0.0433	0.0000	0.0019	0.000	–
galaxies_mean	1000	15743	IsoCal	0.0001	0.0001	0.0001	0.936	0.951
galaxies_mean	1000	15743	LinearCal	0.0002	0.0001	0.0001	0.946	0.980
galaxies_mean	1000	15743	MonoSpline	0.0002	0.0001	0.0001	0.944	0.969
galaxies_mean	1000	15743	Platt	0.0002	0.0001	0.0001	0.942	0.979
galaxies_mean	1000	15743	PPI	0.0002	0.0001	0.0001	0.948	1.000
galaxies_mean	1000	15743	PPI++	-0.0002	0.0001	0.0001	0.946	0.977
galaxies_mean	1000	15743	Venn-Abers	0.0002	0.0001	0.0001	0.942	0.952

Table 2: Representative numerical summary at the smallest, middle, and largest labeled sample sizes for each reproduced benchmark. The reported quantities are Monte Carlo bias, empirical variance of the point estimator, MSE, Wald-interval coverage, and relative efficiency versus PPI for the displayed benchmark set, including PPI++, AIPW-EM, `AutoCal`, `MonoSpline`, and `IsoCal`. Relative efficiency is not reported when the estimator has zero variance.

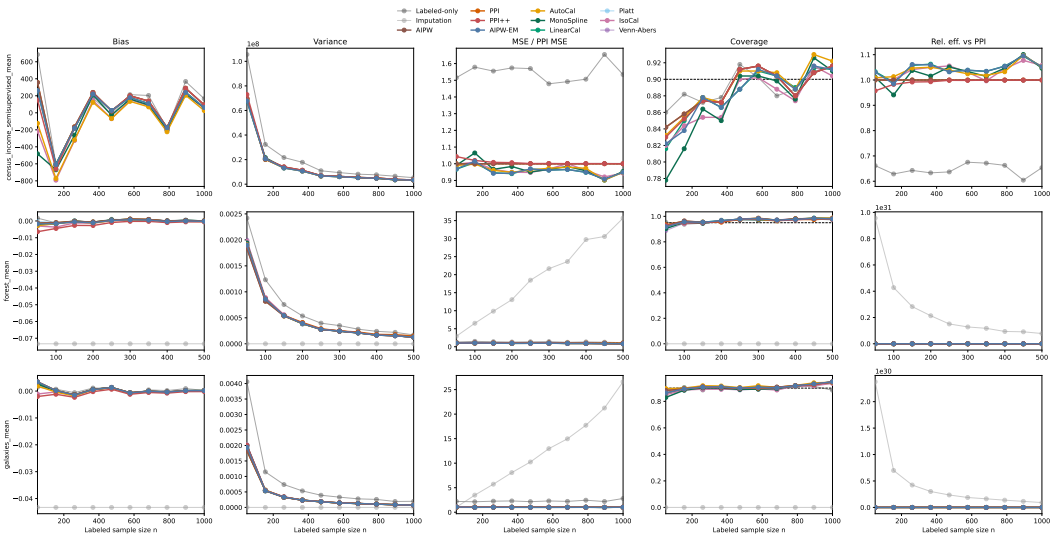


Figure 6: Full diagnostic grid for the reproduced PPI benchmarks, showing bias, empirical variance, normalized MSE relative to PPI, coverage, and relative efficiency versus PPI across labeled-sample-size regimes. In the normalized-MSE and relative-efficiency panels, the dashed horizontal line marks parity with PPI; in the coverage panels it marks the nominal target coverage $1 - \alpha$. `AutoCal`, `MonoSpline`, and `IsoCal` appear alongside PPI, AIPW, PPI++, AIPW-EM, and the fixed calibration rules, while Platt scaling and Venn–Abers appear only on the binary-outcome datasets where they are well defined.

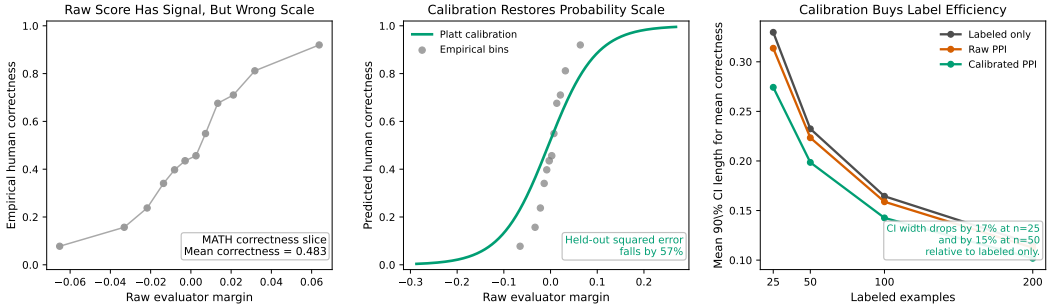


Figure 7: ArmoRM scores and observed MATH correctness. The calibration map is fitted on 100 labeled observations; prediction error is evaluated on the remaining observations.

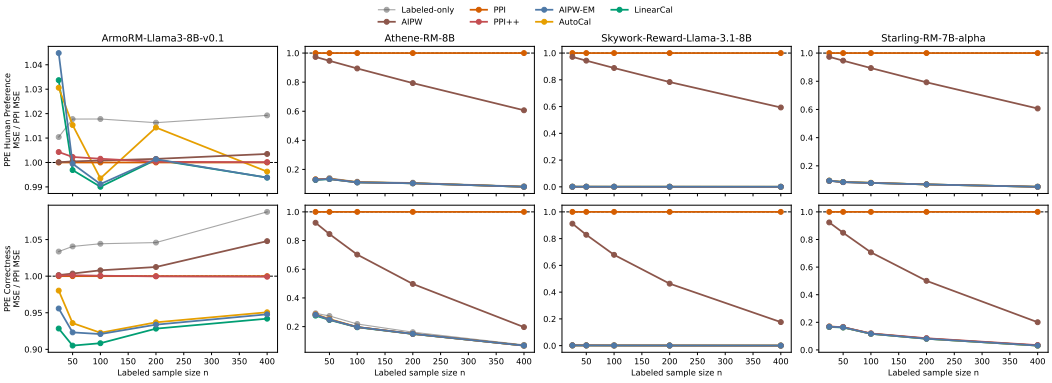


Figure 8: Appendix evaluator-specific PPE summary. Each panel repeats the PPE Human and PPE Correctness comparisons for one public evaluator. The largest MSE increases relative to labeled-only estimation occur for Skywork and Athene.

Track	n	Estimator	MSE / PPI	Label savings	Coverage
PPE Correctness	100	Labeled-only	0.346	0.000	0.927
PPE Correctness	100	PPI	1.000	0.010	0.904
PPE Correctness	100	AIPW	0.774	0.009	0.906
PPE Correctness	100	PPI++	0.330	0.070	0.929
PPE Correctness	100	AIPW-EM	0.309	0.095	0.930
PPE Correctness	100	LinearCal	0.305	0.101	0.931
PPE Correctness	100	AutoCal	0.310	0.094	0.929
PPE Correctness	100	MonoSpline	0.309	0.092	0.926
PPE Correctness	100	IsoCal	0.309	0.089	0.925
PPE Correctness	100	Platt	0.304	0.102	0.930
PPE Correctness	400	Labeled-only	0.299	0.000	0.988
PPE Correctness	400	PPI	1.000	0.020	0.924
PPE Correctness	400	AIPW	0.405	0.009	0.941
PPE Correctness	400	PPI++	0.276	0.083	0.989
PPE Correctness	400	AIPW-EM	0.262	0.103	0.989
PPE Correctness	400	LinearCal	0.260	0.105	0.989
PPE Correctness	400	AutoCal	0.263	0.102	0.988
PPE Correctness	400	MonoSpline	0.263	0.094	0.989
PPE Correctness	400	IsoCal	0.265	0.089	0.989
PPE Correctness	400	Platt	0.260	0.105	0.989
PPE Human Preference	100	Labeled-only	0.303	0.000	0.911
PPE Human Preference	100	PPI	1.000	0.004	0.897
PPE Human Preference	100	AIPW	0.919	0.004	0.896
PPE Human Preference	100	PPI++	0.298	0.027	0.912
PPE Human Preference	100	AIPW-EM	0.295	0.031	0.909
PPE Human Preference	100	LinearCal	0.295	0.031	0.909
PPE Human Preference	100	AutoCal	0.297	0.028	0.908
PPE Human Preference	100	MonoSpline	0.297	0.025	0.903
PPE Human Preference	100	IsoCal	0.297	0.024	0.905
PPE Human Preference	400	Labeled-only	0.289	0.000	0.945
PPE Human Preference	400	PPI	1.000	0.005	0.915
PPE Human Preference	400	AIPW	0.703	0.004	0.917
PPE Human Preference	400	PPI++	0.283	0.025	0.946
PPE Human Preference	400	AIPW-EM	0.282	0.028	0.945
PPE Human Preference	400	LinearCal	0.282	0.028	0.944
PPE Human Preference	400	AutoCal	0.282	0.027	0.944
PPE Human Preference	400	MonoSpline	0.282	0.027	0.942
PPE Human Preference	400	IsoCal	0.282	0.026	0.940

Table 3: PPE numerical summary at $n \in \{100, 400\}$, including the efficiency-maximized and additional fixed-calibration estimators.

n	Estimator	Spearman	Top-1	Top-1 regret
25	AIPW	0.268	0.216	0.064
25	AIPW-EM	0.599	0.344	0.028
25	AutoCal	0.600	0.343	0.028
25	Labeled-only	0.606	0.315	0.030
25	LinearCal	0.599	0.344	0.028
25	PPI	0.265	0.216	0.064
25	PPI++	0.594	0.336	0.029
100	AIPW	0.468	0.285	0.044
100	AIPW-EM	0.845	0.477	0.007
100	AutoCal	0.845	0.487	0.007
100	Labeled-only	0.846	0.485	0.007
100	LinearCal	0.846	0.482	0.007
100	PPI	0.456	0.279	0.046
100	PPI++	0.844	0.479	0.007
400	AIPW	0.631	0.439	0.027
400	AIPW-EM	0.941	0.644	0.002
400	AutoCal	0.941	0.640	0.002
400	Labeled-only	0.940	0.625	0.002
400	LinearCal	0.941	0.646	0.002
400	PPI	0.579	0.419	0.031
400	PPI++	0.940	0.636	0.002

Table 4: Appendix PPE-ranking summary at selected labeled sample sizes from the LLM benchmark. Better rankings have higher Spearman correlation and top-1 identification rates and lower top-1 regret.

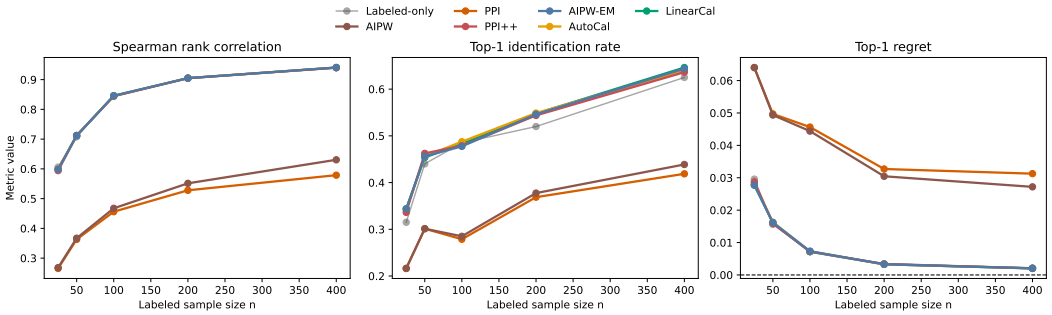


Figure 9: Appendix PPE-ranking supplement from the LLM benchmark, macro-averaged across the public evaluator models. The three panels report Spearman rank correlation, top-1 identification rate, and top-1 regret for recovering the ranking of the eight selected PPE target models from a small labeled sample plus evaluator scores.

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16. **Declaration of LLM usage**

    Question: Does the paper describe the usage of LLMs if it is an important, original, or non-standard component of the core methods in this research?

    Answer: [Yes].

    Justification: LLMs enter only through the empirical LLM-evaluation benchmark, where public precomputed evaluator-score sources and PPE datasets are described in sections 5 and N.3 to N.5; they are not an important, original, or non-standard component of the proposed methodology.

    Guidelines:

    - The answer should be yes, no, or n/a.
    - The paper should describe LLM use that is important, original, or non-standard for the core method, experiments, evaluation, or data generation.
    - The answer n/a means LLMs are not important, original, or non-standard components of the core method.
    - LLM use only for writing, editing, or formatting does not need to be declared when it does not affect methodology, rigor, or originality.